

PAF9615C2: Low Power IR Thermopile Sensor

General Description

The PAF9615C2 is a low-power infrared (IR) thermopile sensor with temperature computing feature. It measures a target object's temperature (T_o) without direct contact with the object. The object temperature is determined by the thermopile sensor and ambient temperature (T_a). The temperatures are converted from the sensors' output voltage and calibrated with built-in calibration coefficients that are saved within the internal memory.

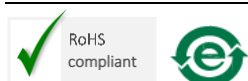
PAF9615C2 supports four output modes of conversion state based on application requirements. The low power consumption advantage makes this thermopile sensor ideal for battery-operated cordless applications.

Key Features

- Integrated MEMS thermopile for non-contact temperature measurement
- 16-bit object temperature data length
 - *Accuracy $\pm 0.5^{\circ}\text{C}$ @ 10 to 45°C (T_o and T_a)
 - Resolution: 0.03125°C
 - FOV: 50°
- 16-bit ambient temperature data length
 - Accuracy $\pm 0.5^{\circ}\text{C}$ @ 10 to 45°C
 - Resolution: 0.03125°C
- Integrated temperature computation function
 - Direct output object temperature
 - Programmable alert temperature
 - Built-in memory for calibration coefficients
- I/O interface
 - Supports two-wire $I^2\text{C}$, 400kbps
 - Supports two programmable $I^2\text{C}$ addresses
- Embedded Emissivity and Decay compensation
- Low power design

Ordering Information

Part Number	Description	Package Type	Packing Type	MOQ
PAF9615C2	Thermopile Sensor	Ceramic	Tray	2,450



For any additional inquiries, please contact us at <https://www.pixart.com>

Preliminary: This product is still under product development where the product specifications are subjected to change upon product release.

Applications

- Non-Contact Object temperature sensing
- Skin temperature monitor to prevent heat stroke
- Ambient temperature detection
- Smart Appliance-Non-contact temperature monitor
- Temperature feedback control application.

Key Parameters

Parameters	Value
Operating Temperature	-20 to 85°C
Object Temperature, T_o	Measurement Range: -20 to 380°C
	*Accuracy $\pm 0.5^{\circ}\text{C}$ @ T_o : 10 to 45°C , T_a : 10 to 45°C
Ambient Temperature, T_a	Measurement Range: -20 to 85°C
	Accuracy: $\pm 0.5^{\circ}\text{C}$ @ 10 to 45°C
Supply Voltage	VDDMA: 2.2 to 3.6 V
	VDDIO: 1.62 to 3.6 V
Typical Current Consumption @ 3.3V, Report Rate = 4Hz	Operation: $225\mu\text{A}$
	Suspend: $2\mu\text{A}$ Shutdown: $2\mu\text{A}$ Note: pure sensor current without I/O toggling. Normal mode @ $t_{\text{CONVERSION}} < t_{\text{REPORT}}$
Package Size	$3.3 \times 3.3 \times 1.53 \text{ mm}^3$

Note: * The higher accuracy could be tuned to optimize for application requiring narrow object temperature range (such as body temperature). For more information, do approach PixArt personnel.

Table of Contents

PAF9615C2: Low Power IR Thermopile Sensor	1
General Description.....	1
Key Features.....	1
Applications.....	1
Key Parameters	1
Ordering Information	1
Table of Contents.....	2
List of Figures.....	5
List of Tables	7
1.0 Introduction	8
1.1 Overview	8
1.2 Block Diagram	8
1.3 Terminology	8
1.4 Signal Description.....	9
2.0 Operating Specifications	10
2.1 Absolute Maximum Ratings	10
2.2 Recommended Operating Conditions	10
2.3 DC Characteristics	11
2.4 AC Characteristics	11
2.5 Performance Specifications	12
2.6 Field of View.....	12
3.0 Mechanical Specifications	13
3.1 Mechanical Dimension	13
3.2 Optical Center Position.....	13
3.3 Package Marking	14
3.4 Packing Information	14
3.4.1 Carrier Drawing.....	14
3.4.2 Unit Orientation.....	15
4.0 Design Reference.....	17
4.1 Reference Schematic.....	17
4.2 PCB Layout Design Guide.....	18
4.2.1 Recommended IC Footprint Layout.....	18
4.2.2 Layout Guideline.....	18
4.2.3 FPC Stiffener Design.....	18
4.3 Measurement Setup.....	19
4.4 Assembly Guide.....	20
4.4.1 Recommended Pb-free Solder Paste.....	20
4.4.2 IR Reflow Soldering Profile	20
4.4.3 Stencil for SMT Process.....	20
4.4.4 Handling Guide	21
5.0 Power Management	22
5.1 Regulators	22
5.2 Power Sequences	22

5.3	Power States	24
5.3.1	States Description	24
5.3.2	State Diagram	24
5.3.3	State Transition.....	25
5.4	Reset	26
5.4.1	Power-on Reset	26
5.4.2	Pin Reset.....	26
5.4.3	Software Reset.....	27
5.5	Shutdown	28
5.6	Related Register	29
6.0	I ² C Serial Interface Communication	30
6.1	Signal Description	30
6.2	Slave ID Selection	30
6.3	Operation Definition.....	30
6.4	Packet Formats.....	32
6.5	Operation Protocol.....	33
6.6	I ² C Timing Specifications.....	35
7.0	System Control	36
7.1	System Initialization	36
7.1.1	Initial Flow	36
7.1.2	Related Register.....	36
7.2	Registers Access	37
7.2.1	Registers Address Mapping.....	37
7.2.2	Register Bank Switching	37
7.2.3	Related Register.....	37
7.3	System Modes.....	37
7.3.1	Normal Mode	39
7.3.2	Interrupt Mode	41
7.3.3	Compare Mode.....	43
7.3.4	One-Shot Mode	45
7.3.5	Related Register.....	46
7.4	Output Access	47
7.4.1	Alert Signals.....	47
7.4.2	Interrupt Method.....	48
7.4.3	Polling Method	49
7.4.4	Data Loss and Corruption.....	50
7.4.5	Related Register.....	51
7.5	Output.....	51
7.5.1	ADC Raw Data Registers.....	51
7.5.2	Temperature Data Registers	52
7.5.3	Related Register.....	52
7.6	Conversion	53
7.6.1	Related Register.....	53
7.7	Report Rate	54
7.7.1	Related Register.....	54

7.8	Filter.....	55
7.8.1	IIR Filter	55
7.8.2	Normal Average Filter	56
7.8.3	Moving Average Filter	57
7.8.4	Related Register.....	57
7.9	Emissivity	57
7.9.1	Related Register.....	57
7.10	Decay	58
7.10.1	Related Register.....	58
7.11	Memory	58
8.0	Register	59
8.1	Register Map.....	59
8.2	Register Description	61
8.2.1	Software Reset Register	61
8.2.2	Product ID Register	61
8.2.3	Bank Switch Register.....	61
8.2.4	System Mode Switching Registers.....	62
8.2.5	Limit Window and Hysteresis Registers.....	62
8.2.6	ADC Raw Data Registers.....	64
8.2.7	Temperature Data Registers	65
8.2.8	Status Registers	66
8.2.9	Sampling Number Registers	66
8.2.10	Report Rate Registers	67
8.2.11	Filter Registers.....	67
8.2.12	Emissivity Register	68
8.2.13	Decay Register	68
8.2.14	ALERT Open Drain Register	68
Revision History		69

List of Figures

Figure 1. Block Diagram	8
Figure 2. Pin Configuration (Bottom View).....	9
Figure 3. FOV Angle	12
Figure 4. Package Mechanical Diagram (unit in mm).....	13
Figure 5. Package Marking	14
Figure 6. Stack 5 +1 JEDEC Tray.....	14
Figure 7. Al Packing Bag	15
Figure 8. Packing Box	15
Figure 9. Orientation	16
Figure 10. Reference Schematic.....	17
Figure 11. Recommend IC Footprint Layout (unit in mm).....	18
Figure 12. FOV Measurement Setup	19
Figure 13. IR Reflow Soldering Profile	20
Figure 14. Power-on Timing Diagram with RSTB_EXT Control	22
Figure 15. Power-on Timing Diagram without RSTB_EXT Control.....	23
Figure 16. Power State Diagram.....	24
Figure 17. Pin Reset Timing Diagram.....	26
Figure 18. Software Reset Timing Diagram	27
Figure 19. Shutdown Timing Diagram	28
Figure 20. START Conditions	30
Figure 21. STOP Conditions.....	31
Figure 22. SDA and SCL Relationship	31
Figure 23. Data Transition Only When SCL Is Low	31
Figure 24. Slave ID Address Packet	32
Figure 25. Register Address Packet	32
Figure 26. Data Packet	32
Figure 27. Single Write.....	33
Figure 28. Single Read	33
Figure 29. Burst Read with Consecutive Addresses	34
Figure 30. I ² C Timing Diagram	35
Figure 31. Initialization Flow	36
Figure 32. Normal Mode.....	39
Figure 33. Interrupt Mode	41
Figure 34. Compare Mode	43
Figure 35. One-Shot Mode.....	45
Figure 36. Regular Data Transmission	47
Figure 37. Interrupt Method.....	48
Figure 38. Polling Method.....	49

Figure 39. Possible Data Loss (1)50

Figure 40. Possible Data Loss (2).....50

Figure 41. Report Rate54

Figure 42. Normal Average56

Figure 43. Moving Average Filter57

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List of Tables

Table 1. Signal Pins Description	9
Table 2. Absolute Maximum Ratings	10
Table 3. Recommended Operating Conditions	10
Table 4. DC Electrical Specifications	11
Table 5. AC Electrical Specifications	11
Table 6. Accuracy Specifications	12
Table 7. Field of View	12
Table 8. Packing Information	14
Table 9. Power On Sequence Timing Specifications	23
Table 10. Power Status	24
Table 11. State Transition	25
Table 12. Pin Reset Sequence Timing Specifications	26
Table 13. Software Reset Sequence Timing Specifications	27
Table 14. Shutdown Sequence Timing Specifications	28
Table 15. I ² C Signals Description	30
Table 16. I ² C ID Address	30
Table 17. I ² C Timing Specifications	35
Table 18. System Modes Registers	37
Table 19. Limit Window and Hysteresis Registers	38
Table 20. Normal Mode Alert Signals	40
Table 21. Interrupt Mode Alert Signals	42
Table 22. Compare Mode Alert Signals	44
Table 23. One-Shot Mode Alert Signals	46
Table 24. Temperature Data Registers	51
Table 25. Conversion Time	53
Table 26. Filter Mode Registers	55
Table 27. IIR Level	55
Table 28. Frame Average Level	56
Table 29. Register Bank 0	59
Table 30. Register Bank 1	60

1.0 Introduction

1.1 Overview

The PAF9615C2 is a low-power infrared (IR) thermopile sensor with temperature computing feature. It measures a target object's temperature (T_o) without direct contact with the object. The object temperature is determined by the thermopile sensor and ambient temperature (T_a). The temperatures are converted from the sensors' output voltage and calibrated with built-in calibration coefficients that are saved within the internal memory. The communication I/O interface is a two-wire I²C (speed up to 400kbps).

PAF9615C2 supports wide object temperature range measurement from -20 to 380°C with 50° FOV. It provides four output modes based on application requirements, including One-Shot, normal, interrupt, and compare mode. The low power consumption advantage makes this thermopile sensor ideal for battery-operated cordless applications.

Note: Throughout this document, the PAF9615C2 is referred to as the “sensor”.

1.2 Block Diagram

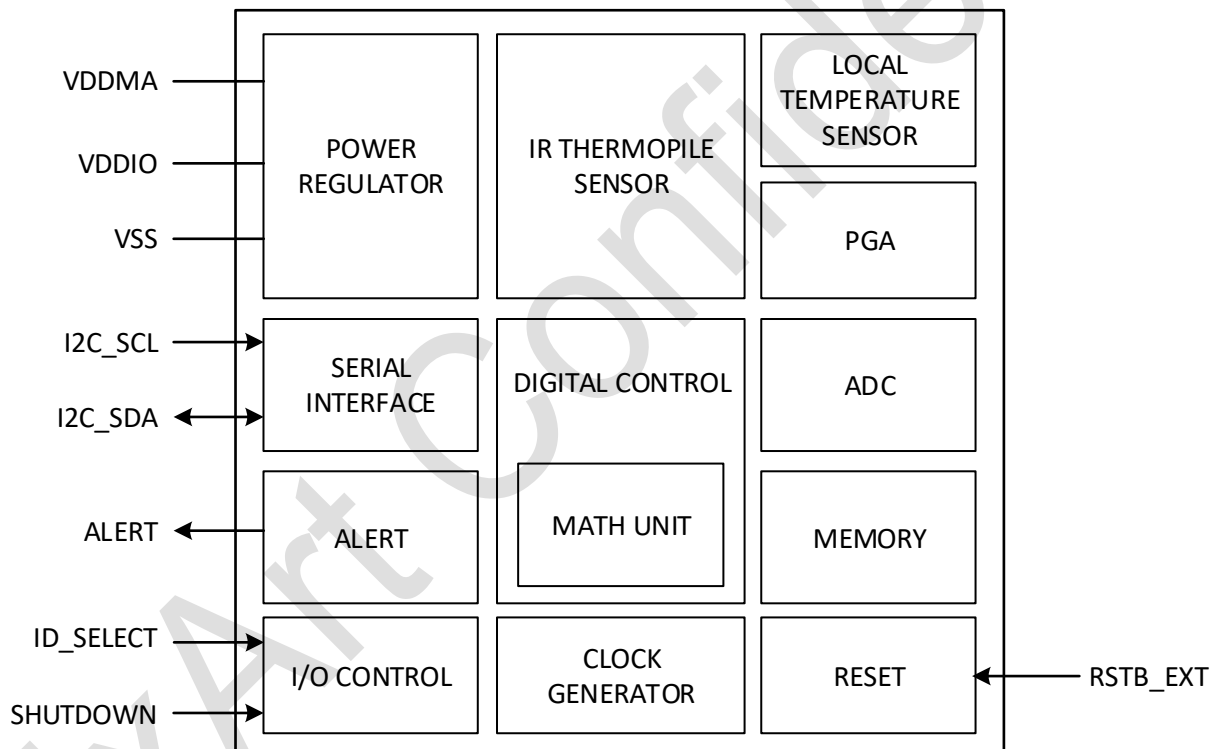


Figure 1. Block Diagram

1.3 Terminology

Term	Description
T_a	Ambient temperature
T_o	Object temperature
FOV	Field of View
SNR	Signal to noise ratio

1.4 Signal Description

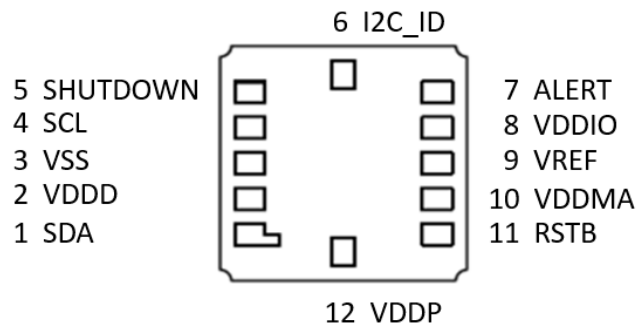


Figure 2. Pin Configuration (Bottom View)

Table 1. Signal Pins Description

Function	Pin No.	Signal Name	Type	Description
Power Supplies	2	VDDD	Power	Internal Regulator output for digital power supply Connects 1 μ F capacitor to Ground
	3	VSS	Ground	Ground
	8	VDDIO	Power	I/O power supply (1.62 to 3.6V). Connects 1 μ F and 0.1 μ F capacitor to Ground
	9	VREF	Power	Analog reference power. Connects 1 μ F capacitor to Ground
	10	VDDMA	Power	Main power supply (2.2 to 3.6V). Connects 1 μ F and 0.1 μ F capacitor to Ground
	12	VDDP	Power	Memory power supply. Connects to VDDD
Interface	1	I2C_SDA	I/O	I ² C bidirectional data pin
	4	I2C_SCL	Input	I ² C clock pin
Functional I/O	5	SHUTDOWN	Input	Shutdown pin; active high High: sensor suspend state and disable I ² C interface Low/ Floating: sensor normal operation
	6	I2C_ID	Input	I ² C 7-bit address selection pin High: I ² C ID= 0x57 Low/ Floating: I ² C ID= 0x34
	7	ALERT	Output	Alert output pin; active low It can be programmed register 0x2F bit[0] as a push-pull or open drain. Push-pull is default
	11	RSTB	Input	Hardware reset pin; active low Floating when not in use

2.0 Operating Specifications

2.1 Absolute Maximum Ratings

Table 2. Absolute Maximum Ratings

Parameter	Symbol	Min.	Max.	Unit	Note
Storage Temperature	T_S	-25	125	°C	
Lead-free Solder Temperature	T_{Peak}	-	260	°C	Refer to Section 4.4 Assembly Guide
Main Power Voltage	VDDMA	-0.4	3.6	V	
I/O Voltage	VDDIO	-0.4	3.6	V	
I/O Pin Voltage	-	-0.4	VDDIO + 0.3	V	All I/O pins
Relative Humidity	RH	0	75	%	Non-condensing, Non-biased
ESD	ESD _{HBM}	-	2	kV	Class 2 on all pins. Human body model, JESD22-A114E.

Notes:

1. At ambient temperature = 25°C.
2. Maximum Ratings are the maximum parameter values that can damage the device when exceeding this limit.
3. Exposure to these conditions or conditions beyond those indicated may adversely affect device reliability.
Functional operation under absolute-maximum-rated conditions is not recommended.

2.2 Recommended Operating Conditions

Table 3. Recommended Operating Conditions

Parameter	Symbol	Min.	Typ.	Max.	Unit	Note
Operating Temperature	T_{OP}	-20	-	85	°C	
Object Temperature	T_o	-20	-	380	°C	
Ambient Temperature	T_a	-20	-	85	°C	
Power Supply Voltage	VDDMA	2.2	3.3	3.6	V	Power regulator input supply. Includes ripples
I/O Supply Voltage	VDDIO	1.62	3.3	3.6	V	Includes ripples
I ² C Speed	SCL_I2C	-	-	400	kHz	Maximum value for Fast mode

Note: PixArt does not guarantee the performance if the operating temperature is beyond the specified limit.

2.3 DC Characteristics

Table 4. DC Electrical Specifications

Parameter	Symbol	Min.	Typ.	Max.	Unit	Note
Digital Supply Voltage	VDDD	1.62	1.8	1.98	V	Internal regulator output. Includes ripples
Analog Reference Voltage	VREF	1.14	1.2	1.26	V	
Power Consumption						
Supply Current @ Operation	I _{DDMA-OPERATION}	-	225	-	μA	t _{CONVERSION} < t _{REPORT} To Number= 2 ¹² Ta Number= 2 ⁹ Report Rate= 4Hz
Supply Current @ Suspend	I _{DDMA-SUSPEND}	-	2	-	μA	Sensor only Suspend without I ² C signal toggling
Peak Power Supply Current	I _{DDMA_MAX}	-	-	255	μA	For VDDMA @ t _{CONVERSION} < t _{REPORT}
I/O						
Input High Voltage	V _{IH}	0.7 x VDDIO	-	-	V	
Input Low Voltage	V _{IL}	-	-	0.3 x VDDIO	V	
Output High Voltage	V _{OH}	VDDIO – 0.4	-	VDDIO + 0.4	V	
Output Low Voltage	V _{OL}	-0.4	-	0.4	V	

Notes: All the parameters are tested under operating conditions: VDDMA= 3.3V, VDDIO= 1.8 and 3.3V, Ta= 25°C.

2.4 AC Characteristics

Table 5. AC Electrical Specifications

Parameter	Symbol	Min.	Typ.	Max.	Unit	Note
Boot Up Duration	t _{PU}	120	-	-	ms	From VDDMA rising to valid interface communication

Notes: All the parameters are tested under operating conditions: Ta= 25°C, VDDMA= 3.3V, VDDIO= 3.3V for 3.3V I/O application and VDDIO= 1.8 V for 1.8V I/O application.

2.5 Performance Specifications

All the sensor unit is calibrated before delivery. Both calibration and thermal transient coefficient data are saved in integrated memory.

Table 6. Accuracy Specifications

Parameter	Symbol	Min.	Typ.	Max.	Unit	Note
T_object (To) Accuracy			*±0.5		°C	To= 10 to 45°C and Ta= 10 to 45°C
T_ambient (Ta) Accuracy			±0.5		°C	Ta= 10 to 45°C
			±1		°C	Ta= 0 to 70°C

The sensor can be optimized to support specific application through compromising between object temperature range and accuracy requirement. (For example, human body temperature.). For further enquiry, do approach PixArt personnel.

The Table 6 specified accuracy are under below condition.

- Under the isothermal conditions
- At VDDMA= 3.3V
- The target object needs to be larger than FOV
- The emissivity of the object is 0.98
- Ta and To sampling numbers are set to 512 and 4096, refer to Section 7.6 for details

2.6 Field of View

Table 7. Field of View

Parameter	Symbol	Min.	Typ.	Max.	Unit	Note
FOV angle		40	50	60	Deg	50% of Maximum

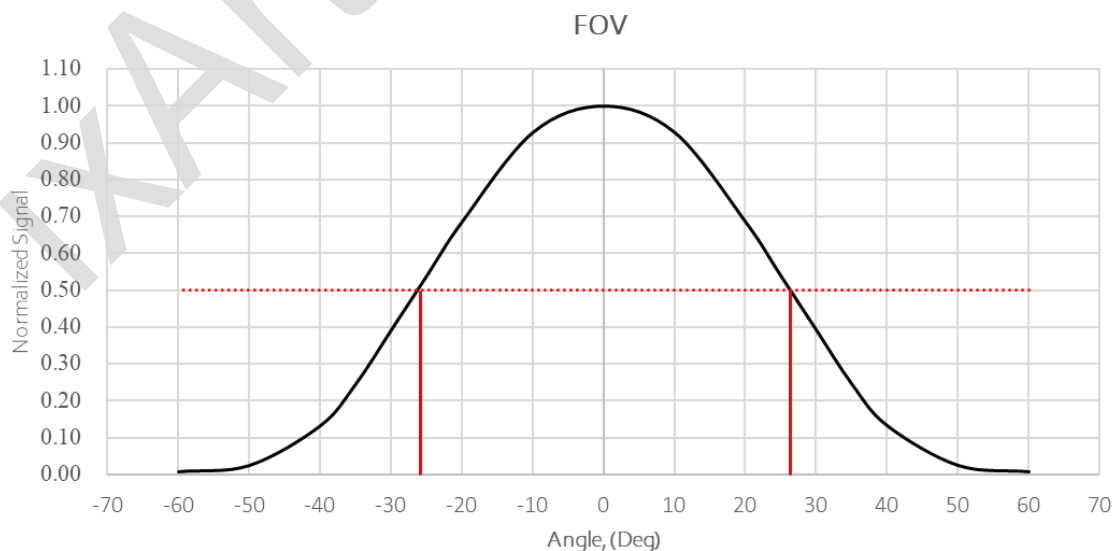


Figure 3. FOV Angle

3.0 Mechanical Specifications

3.1 Mechanical Dimension

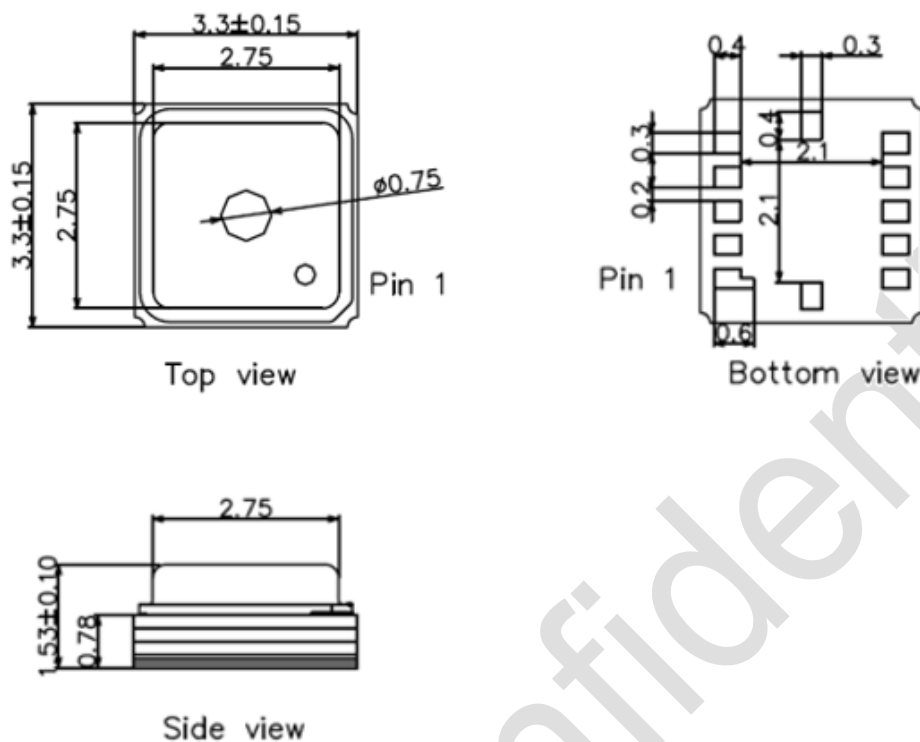


Figure 4. Package Mechanical Diagram (unit in mm)

3.2 Optical Center Position

Optical center position is same as sensor center position.

3.3 Package Marking

Refer to Figure 5. Package marking for the code marking location on the sensor package.

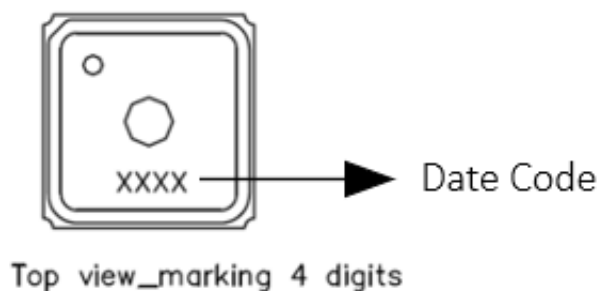


Figure 5. Package Marking

3.4 Packing Information

3.4.1 Carrier Drawing

Table 8. Packing Information

Parameter	Quantity	Unit	Note
Sensors per tray	490	piece	
Trays per stack	5	tray	Added a top cover on top of each stack. Refer to Figure 6
Stacks per packing bag	1	stack	Refer to Figure 7
Packing bags per packing box	1	bag	Refer to Figure 8
Max. sensors per packing box	2450	piece	



Figure 6. Stack 5 +1 JEDEC Tray



Figure 7. Al Packing Bag



Figure 8. Packing Box

3.4.2 Unit Orientation

Packing method of using JEDEC Tray. CLCC Pin 1 orientation towards the edge of the JEDEC tray chamfer location.

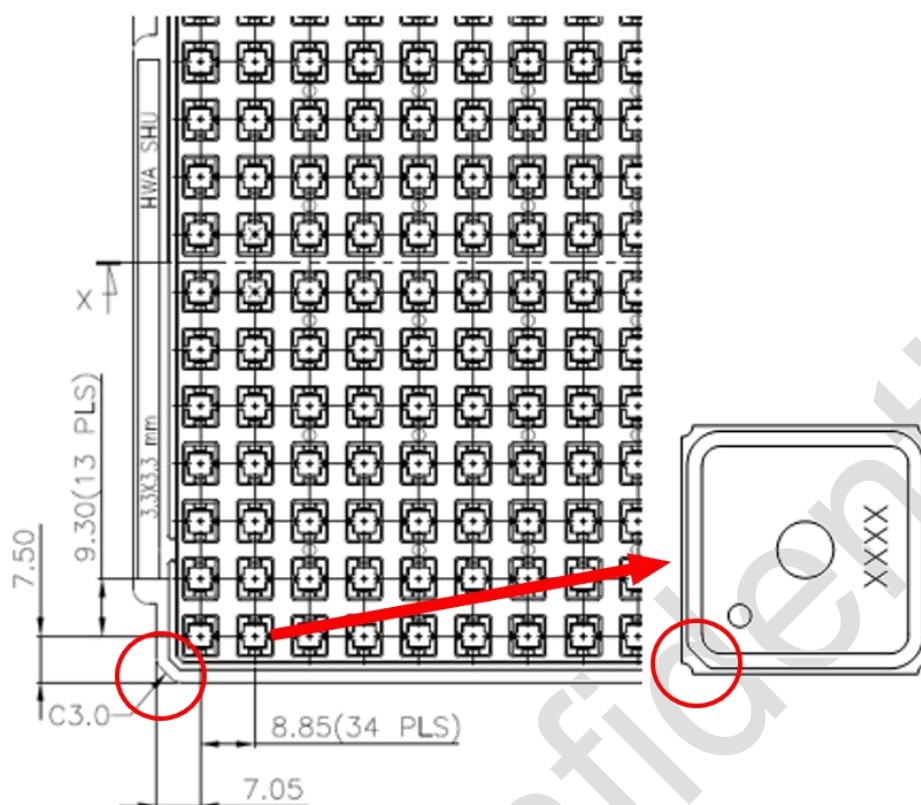


Figure 9. Orientation

4.0 Design Reference

4.1 Reference Schematic

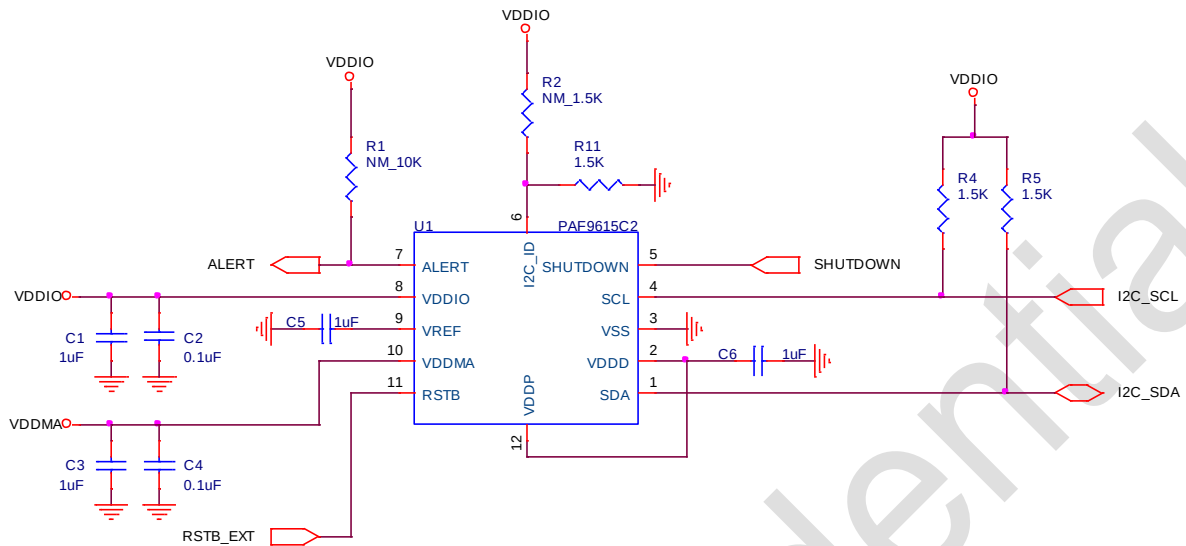


Figure 10. Reference Schematic

Notes:

1. VDDMA= 2.2 to 3.6V, VDDIO= 3.3V for 3.3V system.
2. VDDMA= 2.2 to 3.6V, VDDIO= 1.8V for 1.8V system.
3. I2C_SDA and I2C_SCL are open drain output and must pull up to VDDIO.
4. Alert pin can be programmed as a push-pull or open drain (push-pull is the default setting) and recommend connect to the MCU GPIO as data ready for power saving.
5. VDDMA and VDDIO connect 1 μ F and 0.1 μ F capacitor to the ground and place close to the sensor.
6. VDDD and VREF connect 1 μ F capacitor to the ground respectively and place close to the sensor.
7. For this reference, I2C_ID address is setting 0x34 and sensor in normal operation (shutdown pin is setting to low).

4.2 PCB Layout Design Guide

4.2.1 Recommended IC Footprint Layout

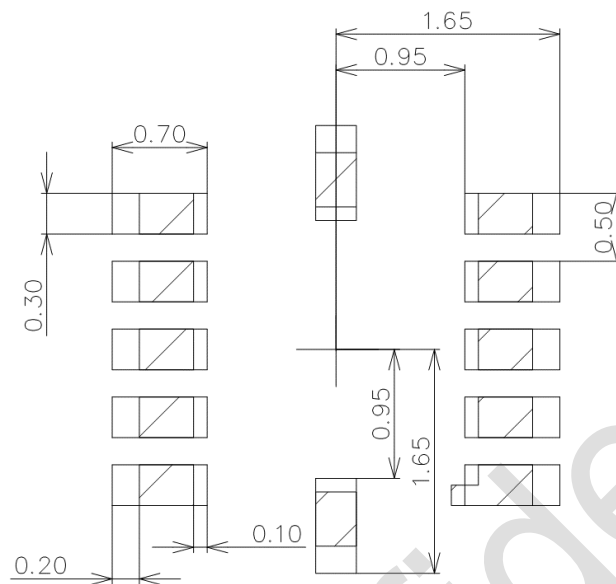


Figure 11. Recommend IC Footprint Layout (unit in mm)

4.2.2 Layout Guideline

1. Minimum 10 mils traces width for VDDMA, VDDIO, VDDD, VDDP, VREF and VSS.
2. Place all of capacitors close to the sensor, especially capacitor of VREF pin.
3. Avoid any heat sources from any layer (e.g., MCU, power source, power trace, etc.).
4. A solid ground plane under the sensor is recommended.

4.2.3 FPC Stiffener Design

For FPC board design, a stiffener at the FPC bottom side is required to strengthen the board.

Recommended stiffener dimension:

1. FR4 with minimum thickness of 0.4mm.
2. Stainless steel with minimum thickness 0.15mm.

4.3 Measurement Setup

1. No object is allowed in FOV area between target and sensor.
2. Target object must be larger than FOV area.
3. Optical center position is same as sensor center position. Target object center position must align to sensor center.

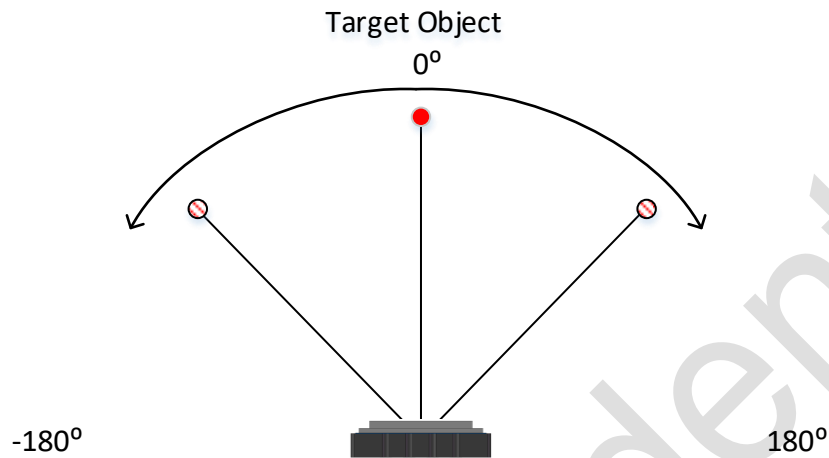


Figure 12. FOV Measurement Setup

4.4 Assembly Guide

4.4.1 Recommended Pb-free Solder Paste

1. Almit LFM-48W TM-HP
2. Senju M705-GRN360-K

4.4.2 IR Reflow Soldering Profile

Temperature profile is used for reflow soldering that must be fine-tuned to establish a robust process. The typical recommended IR reflow profile as shown below.

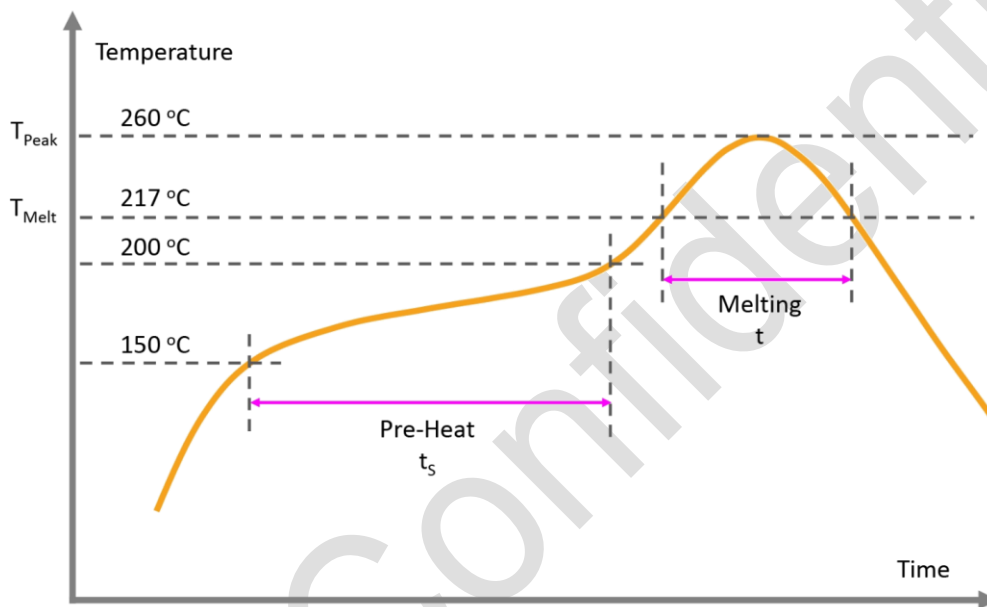


Figure 13. IR Reflow Soldering Profile

Parameter	Symbol	Min.	Max.	Unit	Note
Ramp-up rate	T _{RAMP}		3	°C/sec	From T _{Melt} to T _{Peak}
Pre-heat area temperature	T _{PRE}	150	200	°C	
Pre-heat area duration	t _s	60	120	sec	
Melting duration	t	60	150	sec	
Melting Temperature	T _{Melt}	217		°C	
Peak Temperature	T _{Peak}		260	°C	

4.4.3 Stencil for SMT Process

Recommended SMT process, stencil thickness is from 0.1 to 0.12 mm.

1. The stencil thickness selection will need to consider passive component around the sensor.
2. For surface mounted sensor on FPC, the cover layer and adhesive layer total thickness recommended to be less than 40 μm.
3. The recommended ratio of the stencil opening to the pad size is 1:1.

4.4.4 Handling Guide

Do not use air gun to blow on the sensor throughout the handling process.



5.0 Power Management

5.1 Regulators

The sensor has integrated regulators for internal power supply. VDDD is the internal regulator terminal pad to connect 1 μ F capacitor to ground for voltage stabilization.

5.2 Power Sequences

- Power On sequence with RSTB_EXT control
 - VDDMA is required to Power on first.
 - VDDIO power up timing must be equal or later than VDDMA.
 - RSTN_EXT must always be asserted when the Power on in progress.
 - SHUTDOWN must tie Low before RSTN_EXT asserted if used SHUTDOWN function.
 - VDDMA to RSTN_EXT interval must be at least 1ms.
 - Wait for at least 120ms after RSTN_EXT de-asserted, the sensor enters suspend state for initial process.

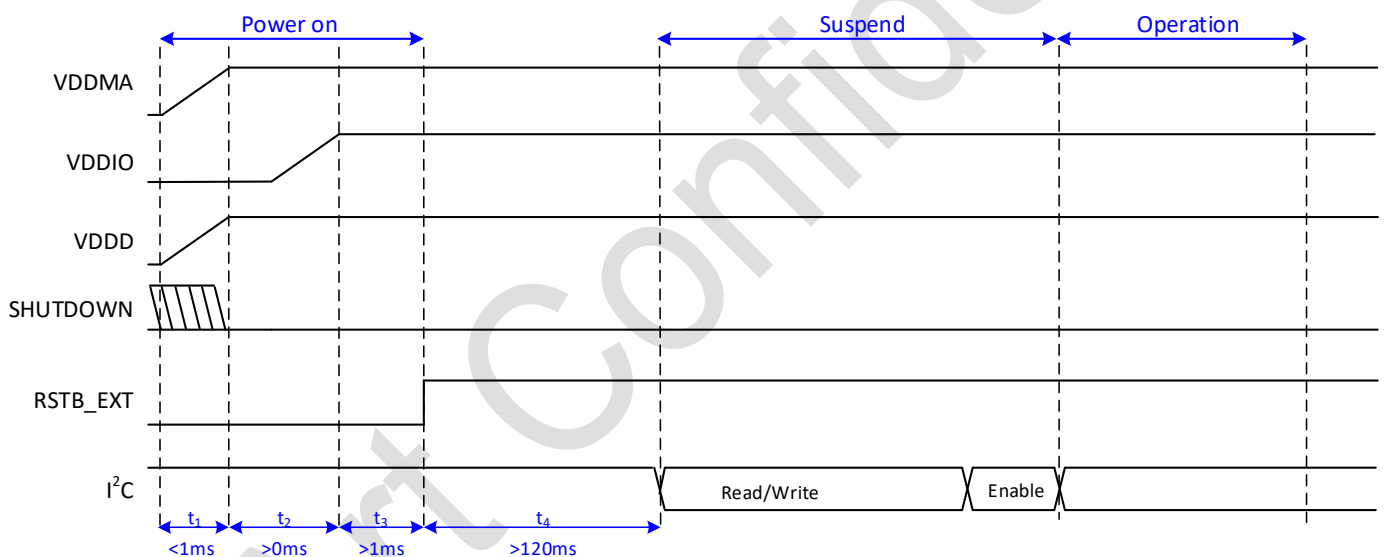


Figure 14. Power-on Timing Diagram with RSTB_EXT Control

- Power On sequence without RSTB_EXT control
 - VDDMA is required to Power on first.
 - VDDIO power up timing must be equal or later than VDDMA.
 - SHUTDOWN must tie Low before VDDIO power on if used SHUTDOWN function.
 - Wait for at least 120ms after Power on is completed, the sensor enters suspend state for initial process.

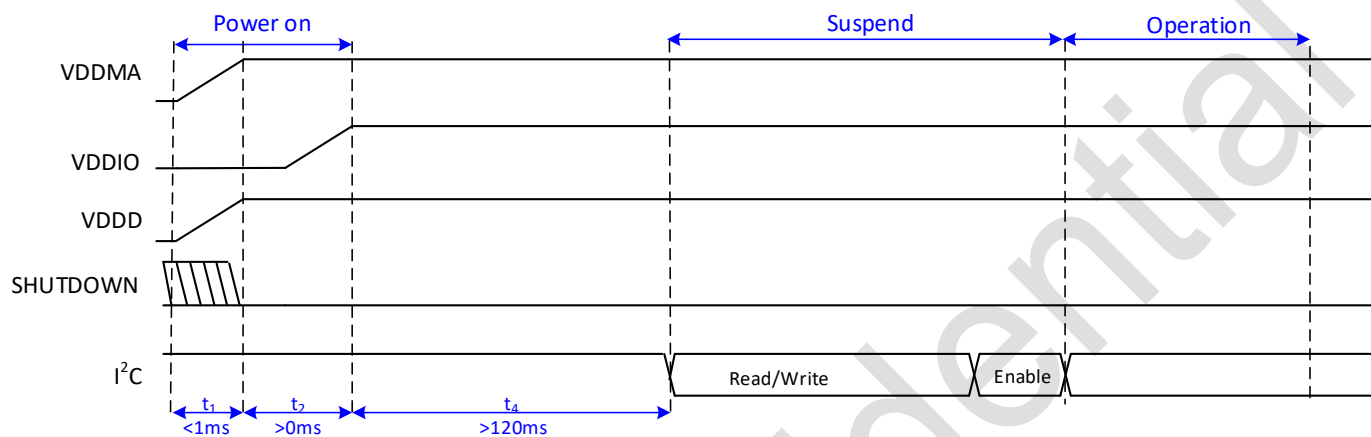


Figure 15. Power-on Timing Diagram without RSTB_EXT Control

Table 9. Power On Sequence Timing Specifications

Parameter	Label	Min.	Max.	Unit
VDDMA/ VDDD rise time	t ₁	-	1	ms
VDDMA rising to VDDIO rising	t ₂	0	-	ms
VDDIO rising to RSTB_EXT rising	t ₃	1	-	ms
RSTB_EXT rising to Suspend	t ₄	120	-	ms

5.3 Power States

5.3.1 States Description

Table 10. Power Status

State	Description
Off	No power supply, all the voltage rails and clocks are gated.
Suspend	Upon powered on, the sensor enters Suspend state mainly for configuring the initial and customized sensor parameter settings. The sensor is not activated for power saving until host command is fully executed.
Shutdown	The sensor enters shutdown state and I ² C bus is disabled for power saving. The typical current consumption in shutdown state is less than 2μA. The power supply VDDMA, VDDIO and VDDD is remain power-on.
Operation	This is the state where the sensor repeatedly measuring and performing the conversion of To/Ta data.

5.3.2 State Diagram

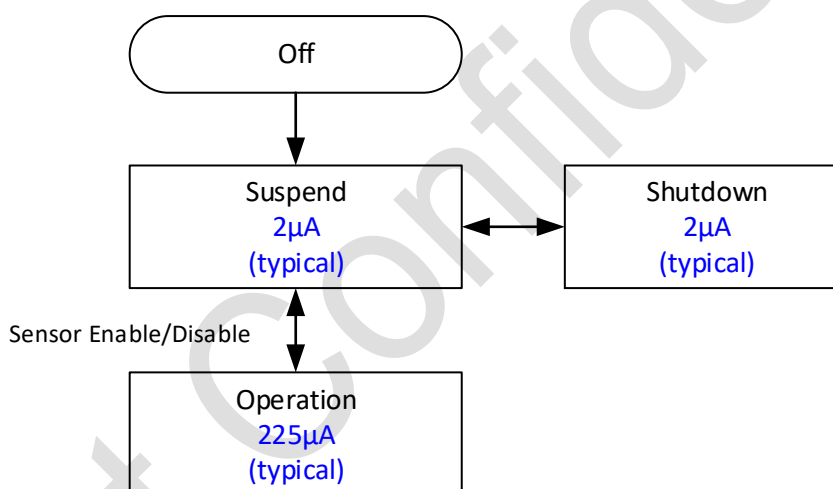


Figure 16. Power State Diagram

5.3.3 State Transition

Table 11. State Transition

State	Type	Description
Off to Suspend	Power on	The sensor hardware complete power on process. The sensor I ² C interface is ready for the host communication upon initialization complete. The sensor is available to receive command from the host.
Suspend to Shutdown	I/O	The sensor transition from Suspend state to the Shutdown by Shutdown pin control. When the sensor shutdown enabled (shutdown pin asserted), the sensor enters shutdown state and I ² C bus is disabled for power saving.
Shutdown to Suspend	I/O	When shutdown pin is de-asserted, the sensor exits from shutdown state to Suspend state. The registers value is remained intact during shutdown pin is asserted.
Suspend to Operation	Command	The sensor transition to Operation state from Suspend by the sensor enable command.
Operation to Suspend	Command	For Normal, Interrupt and Compare Mode. The sensor transition from Operation state to Suspend by the sensor disable command.
	Automation	One-Shot mode only As soon as the sensor completed the One-Shot temperature conversion, the sensor will return to suspend state automatically.

- Transition from Suspend to Operation State (sensor enable):

1. Write register 0x7F with value 0x00
2. Write register 0x04 with value 0x01

- Transition from Operation to Suspend State (sensor disable):

1. Write register 0x7F with value 0x00
2. Write register 0x04 with value 0x00

5.4 Reset

There are three approaches to trigger for sensor reset. The sensor needs to re-initialize due to all previous registers will be cleared after each reset except software warm reset.

5.4.1 Power-on Reset

During power on, the sensor does not need an external reset trigger as there is an internal circuitry that performs power-on reset function for the sensor.

5.4.2 Pin Reset

The sensor can be reset by pulling RSTB_EXT pin. When RSTB_EXT pin is asserted (pull down), the sensor will be reset. Wait for 120ms after the RSTB_EXT de-asserted, the sensor transition back to suspend state.

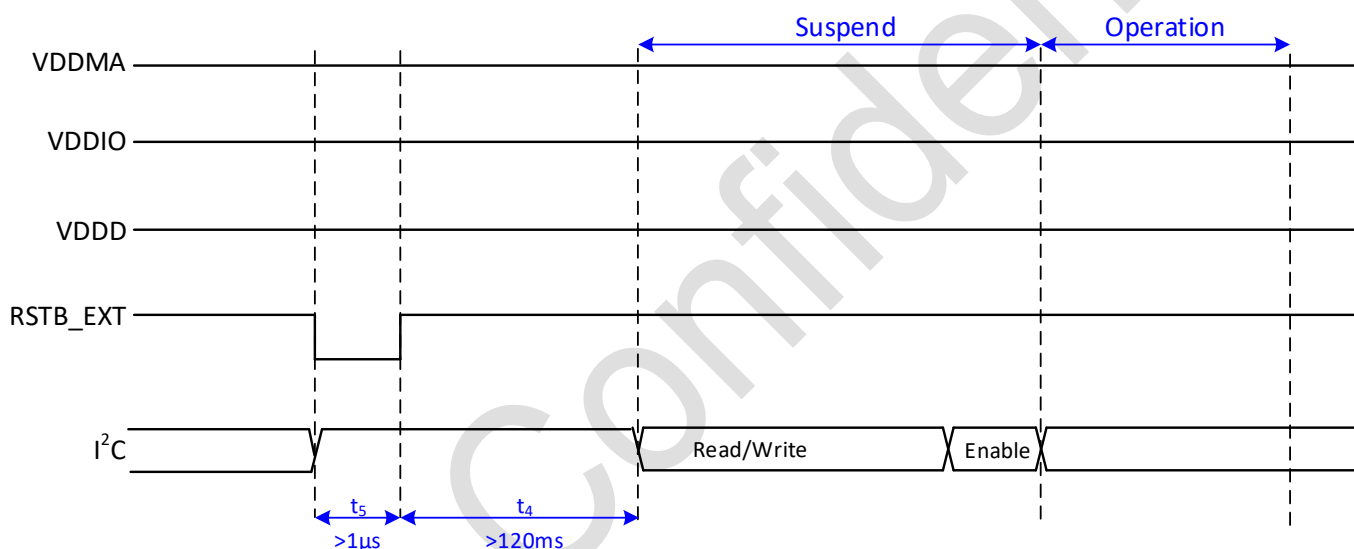


Figure 17. Pin Reset Timing Diagram

Table 12. Pin Reset Sequence Timing Specifications

Parameter	Label	Min.	Max.	Unit
RSTB_EXT assert hold time (pull down)	t_5	1	-	μs
RSTB_EXT rising to Suspend	t_4	120	-	ms

5.4.3 Software Reset

The sensor can be reset by setting the *Host_SW_Reset* register Bank 0, address 0x7D).

There are two types of software for this sensor:

- Cold reset – set register value to 0x5A
All digital functions will be reset and register settings need to be reinitialized. Wait for 120ms, then sensor transition back to suspend state.
- Warm reset – set register value to 0x9A
All digital functions will be reset except the register settings and wait for 120ms. Then, the sensor transition back to suspend state.

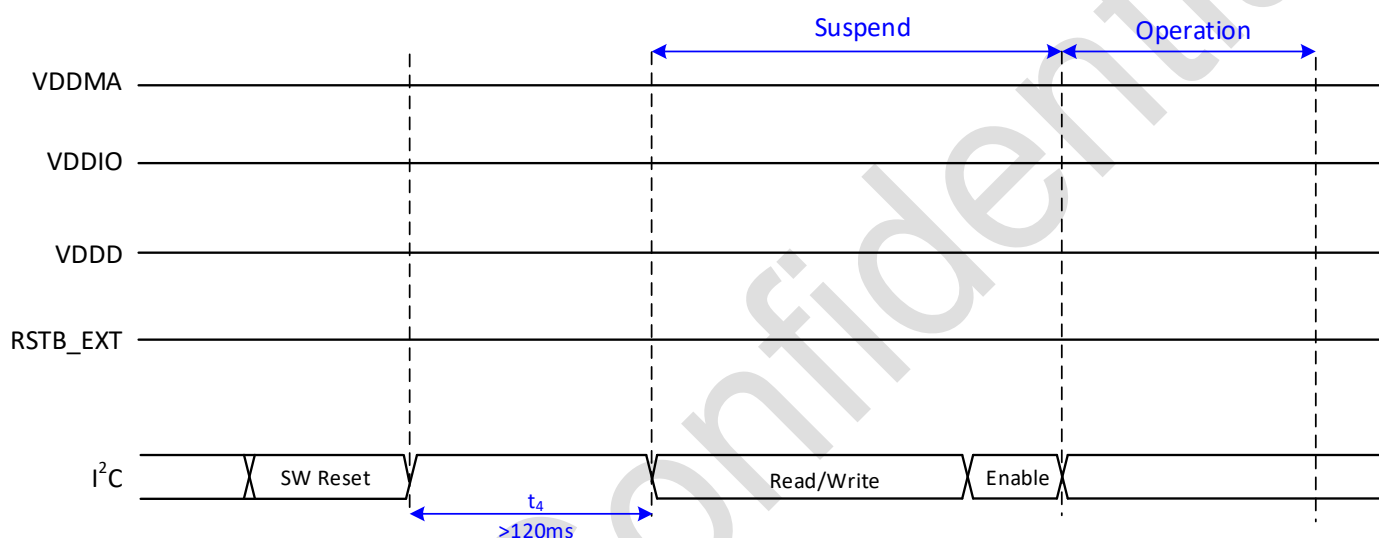


Figure 18. Software Reset Timing Diagram

Table 13. Software Reset Sequence Timing Specifications

Parameter	Label	Min.	Max.	Unit
Software Reset to Suspend State	t ₄	120	-	ms

5.5 Shutdown

The Shutdown is I/O control pin for sensor enters Shutdown state and exit from shutdown state. When the sensor shutdown enabled (shutdown pin asserted), the sensor enters shutdown state and I²C bus is disabled for power saving. The typical current consumption in shutdown state is less than 2μA. The power supply VDDMA, VDDIO and VDDD is remain power-on. When shutdown pin is de-asserted, the sensor exits from shutdown state to Suspend state. The registers value is remained intact during shutdown pin is asserted.

There are three states for shutdown pin:

- Shutdown pin is High
Sensor enters suspend state with disable I²C interface (I2C_SDA/SCL always keep high level).
- Shutdown pin is Low
Sensor normal operation.
- Shutdown pin is floating
When shutdown pin is floating, the shutdown pin is pull down by the internal pull-down resistor. Sensor normal operation.

The following are the scenarios where the Shutdown pin control is prohibited.

- When VDDMA Power on process is in progress
- When sensor is in operation state

During shutdown state, if the host issues a hardware pin reset, the sensor will reset immediately. The host needs to de-assert the shutdown pin and reinitialize the sensor.

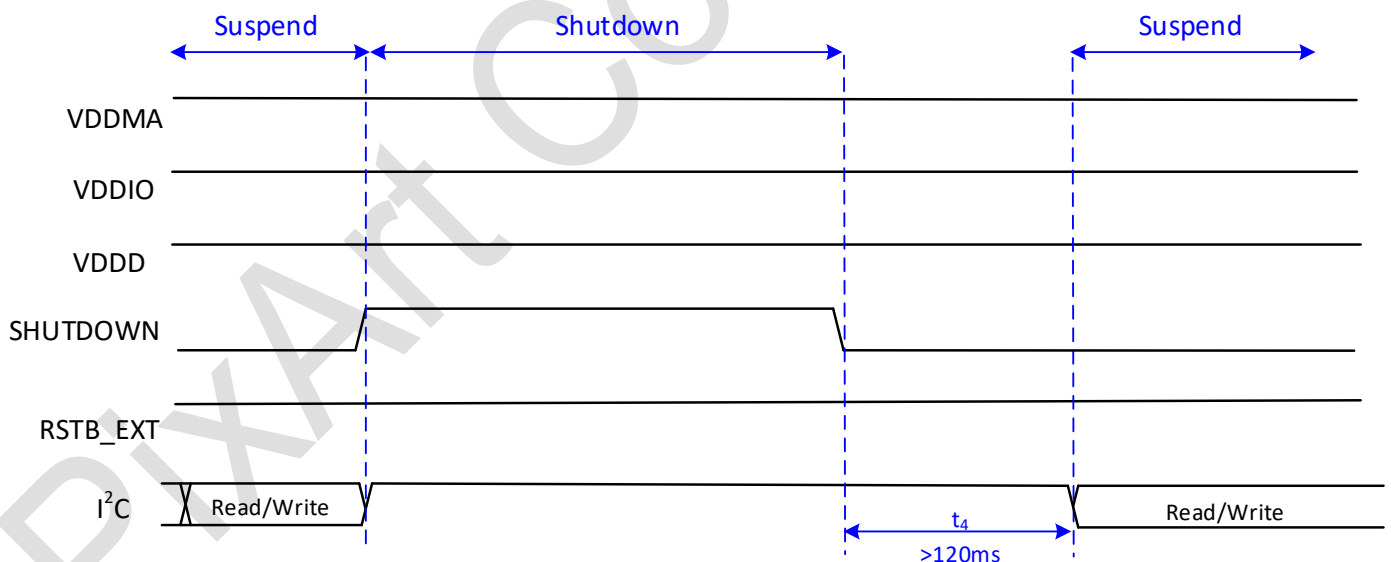


Figure 19. Shutdown Timing Diagram

Table 14. Shutdown Sequence Timing Specifications

Parameter	Label	Min.	Max.	Unit
Shutdown Falling to Suspend	t ₄	120	-	ms

5.6 Related Register

Usage	Name	Bank	Address
Software Reset	<i>Host_SW_Reset[7:0]</i>	0x00	0x7D

6.0 I²C Serial Interface Communication

6.1 Signal Description

This sensor uses two wires synchronous I²C interface. The two signals are SCL (clock) and SDA (data). The sensor is design as slave only device, drives SDA read cycles and issue acknowledgement bit for confirmation. The sensor uses 7-bit addressing and does not support clock stretching.

Table 15. I²C Signals Description

Signal Name	Type	Reset State	Description
I2C_SCL	Input	Input	The SCL is the I ² C serial clock pin and driven by the host where is to synchronize the serial data transmission. A pull-up resistor is required.
I2C_SDA	I/O	Input	The SDA signal is data line to read from or write to the sensor. A pull-up resistor is required.

6.2 Slave ID Selection

The sensor uses 7-bit addressing and the slave ID is decided by the status of I2C_ID pin. The I2C_ID pin is recommended to be not connected (floating), tie to Ground, or pull up to VDDIO instead host control directly. When I2C_ID pin floating, the pin will be pull-down by sensor's internal pull-down resistor. When finishing of power-on sequence processes, I2C_ID pin state change is prohibited.

Table 16. I²C ID Address

I2C_ID Status	ID Address	Description
High	0x57	Tie to VDDIO with a pull up resistor.
Low	0x34	Tie to Ground.
Floating	0x34	Pin floating. No Connection.

6.3 Operation Definition

All transactions begin with a START (S) and are terminated by a STOP (P). START and STOP conditions is always generate by the host.

START condition: A HIGH to LOW transition on the SDA line while SCL is HIGH.

STOP condition: A LOW to HIGH transition on the SDA line while SCL is HIGH.

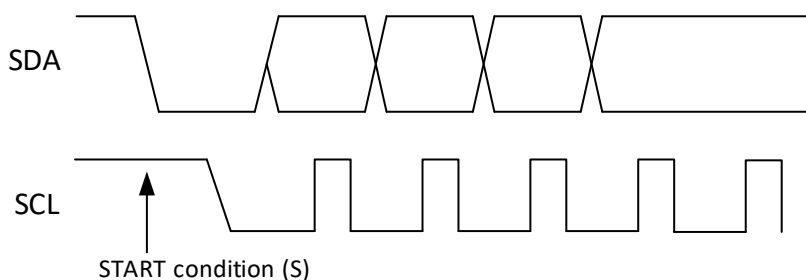


Figure 20. START Conditions

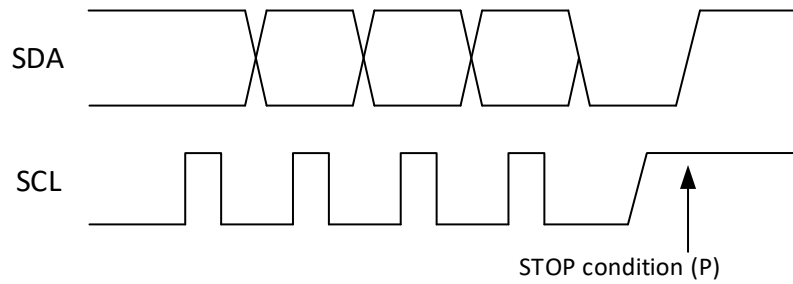


Figure 21. STOP Conditions

For a complete transmission, total 9 clock cycle is required. First 8 clock cycles are a complete one-byte data transmission and the 9th clock cycle is for acknowledgement (0= ACK) or not acknowledge (1= NACK) confirmation. Refer to Figure 22 for detail.

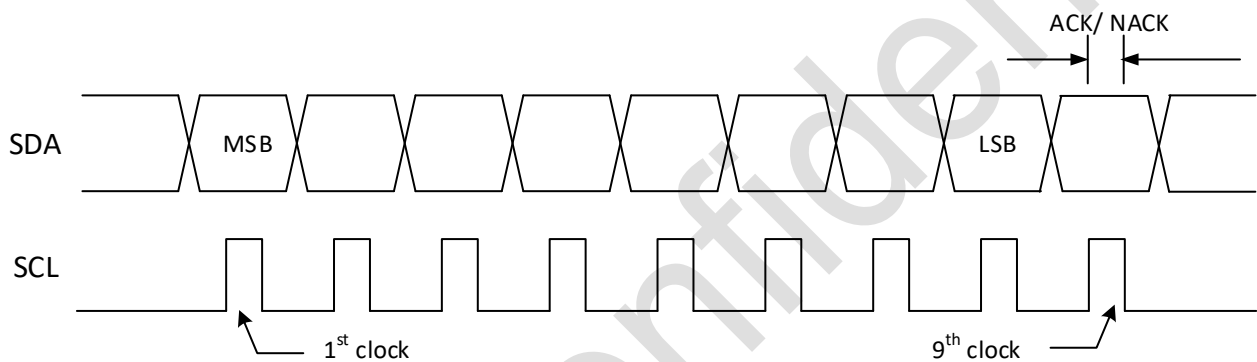


Figure 22. SDA and SCL Relationship

During transmission, the SDA transition when SCL at low state.

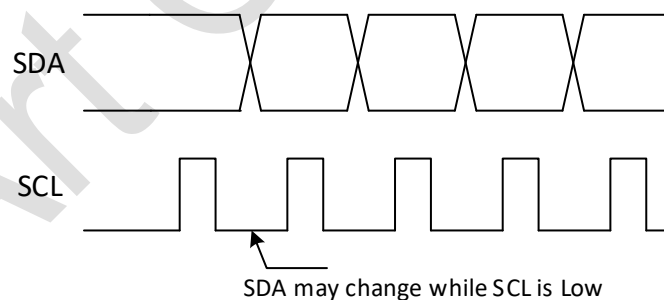


Figure 23. Data Transition Only When SCL Is Low

The host controls SDA transmission in three different conditions, when one of the three condition is fulfilled.

- When sending the Slave ID and register address to the sensor.
- When writing register data.
- After received a byte of data from the sensor, the host will update an ACK or NACK bit.

The sensor controls the SDA under two conditions, when one of the two condition is fulfilled.

- When sending data to the host.
- After received a byte of data from the host, the sensor will update an ACK bit to confirm.

6.4 Packet Formats

A transmission packet comprises three different formats. All packets are eight bits or one byte long.

- Slave ID Address Packet: Command packets contain a 7-bit device address and a read/write bit (R/W)

Read/Write bit:

The Read/Write bit defines the direction of all data bytes after the Start condition (Read: 1, Write: 0). System allows one operation (either Read or Write) at a time.

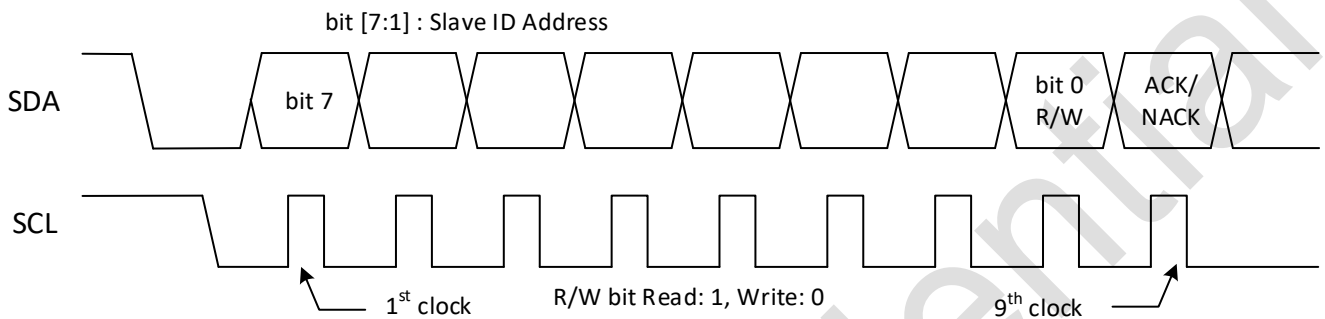


Figure 24. Slave ID Address Packet

- Register Address Packet:

The address packets contain a 7-bit register address with first bit as address mode bit. The address mode bit defines the address incremental option for multi-register data access.

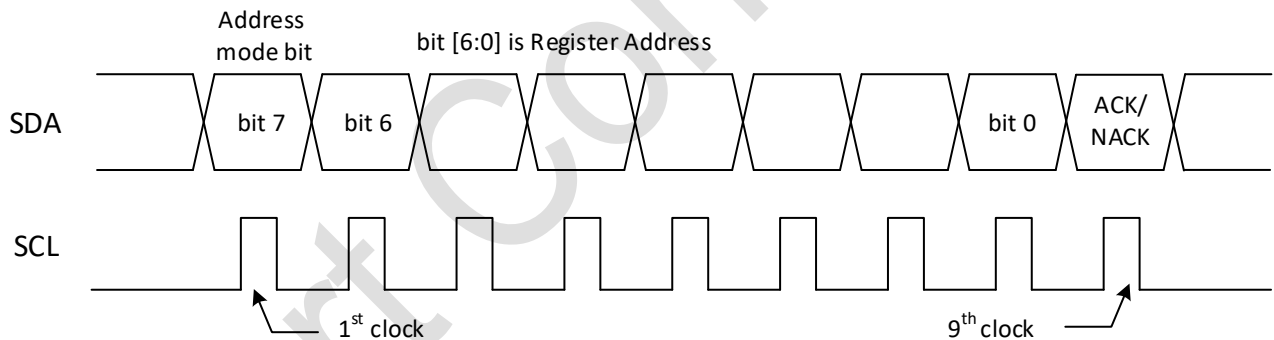


Figure 25. Register Address Packet

- Data Packet:

Contains 8-bit data either send by the host or read from the sensor register.

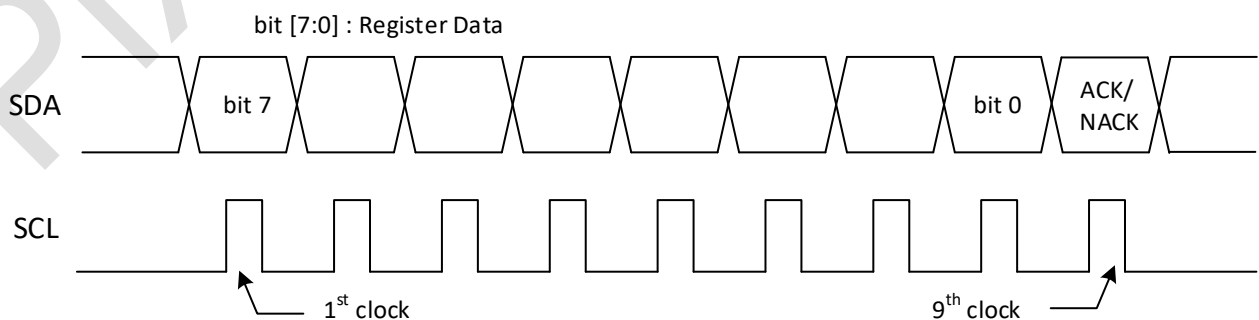


Figure 26. Data Packet

6.5 Operation Protocol

The protocol is derived from the three above-mentioned packet formats. Below is the combination of the packet format in different operations.

Single Write:

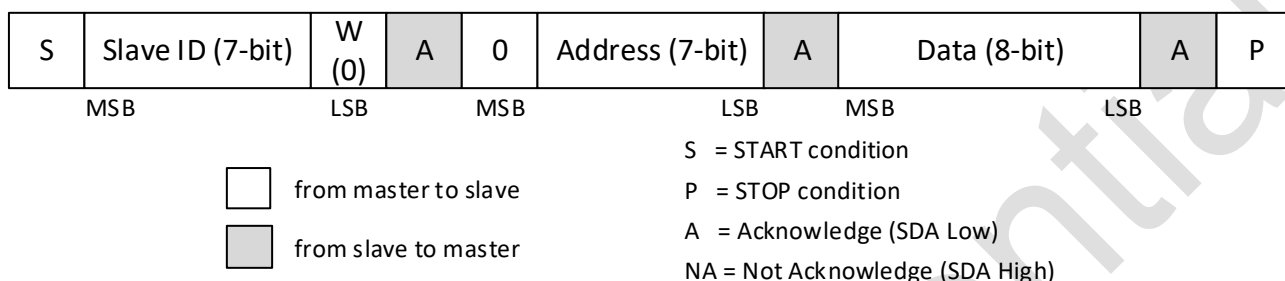


Figure 27. Single Write

1. The host initiates a START condition with Slave ID and LSB bit is '0' as write.
2. If matches Slave ID address, the sensor responds an ACK.
3. The host sends register address.
4. The sensor responds an ACK when receiving address command.
5. The host sends data to the sensor.
6. The sensor writes data to register and responds an ACK.
7. The host terminates the operation with a STOP condition.

Single Read:

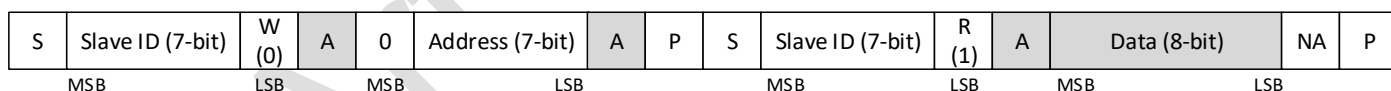


Figure 28. Single Read

1. The host initiates a START condition with Slave ID and LSB bit is '0' as write.
2. If matches Slave ID address, the sensor responds an ACK.
3. The host sends register address.
4. The sensor responds an ACK upon receiving address command.
5. The host terminates the operation with a STOP condition.
6. The host reinitiates a START condition with Slave ID and LSB bit is '1' as read.
7. If matches Slave ID address, the sensor responds an ACK.
8. The sensor sends register data to the host.
9. The host responds a NACK and terminate the operation with a STOP condition.

- Burst Read with Consecutive Addressing:



Figure 29. Burst Read with Consecutive Addresses

When the address mode is 0, the register address will auto increase by 1 and output data in sequence:

1. The host initiates a START condition with Slave ID and LSB bit is '0' as write.
2. If matches Slave ID address, the sensor responds an ACK.
3. The host sends register address with MSB bit is '0' in address mode bit.
4. The sensor responds an ACK upon receiving address command.
5. The host terminates the operation with a STOP condition.
6. The host reinitiates a START condition with Slave ID packet LSB bit is '1' as read.
7. If matches Slave ID address, the sensor responds an ACK.
8. The sensor sends register data to the host with register address.
9. The host continues issue an ACK for next register data read request with register address incremental by 1.
10. The host repeats step 9 until a NACK is issued.
11. The host terminates the operation with a STOP condition.

6.6 I²C Timing Specifications

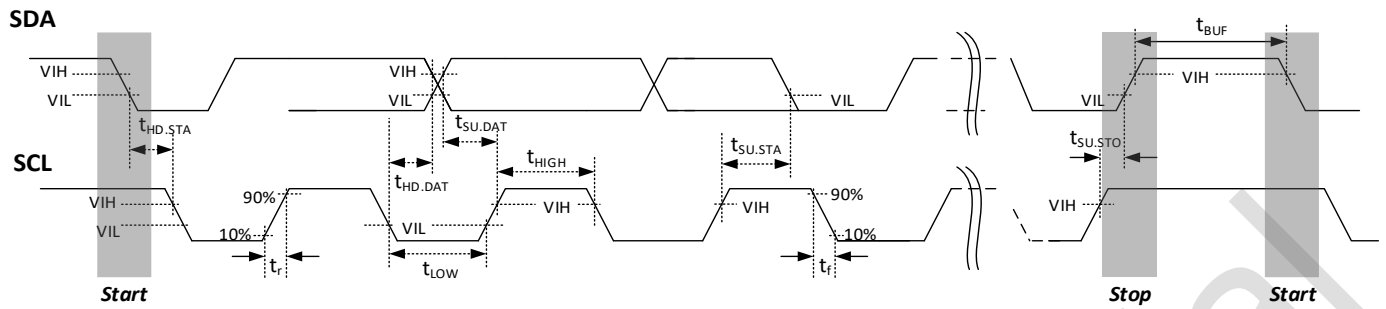


Figure 30. I²C Timing Diagram

Table 17. I²C Timing Specifications

Parameter	Symbol	Standard Mode		Fast Mode		Unit
		Min.	Max.	Min.	Max.	
SCL clock frequency.	f_{scl}		100		400	kHz
Hold time for Start/Repeat Start. After this period, the first clock pulse is generated.	$t_{HD.STA}$	4		0.6		μs
Set-up time for a repeated Start.	$t_{SU.STA}$	4.7		0.6		μs
Low period of SCL clock.	t_{LOW}	4.7		1.3		μs
High period of SCL clock.	t_{HIGH}	4		0.6		μs
Data hold time.	$t_{HD.DAT}$	0		0		μs
Data set-up time.	$t_{SU.DAT}$	250		100		ns
Rise time of both SDA and SCL signals.	t_r		1000	20	300	ns
Fall time of both SDA and SCL signals.	t_f		300	-	300	ns
Set-up time for STOP condition.	$t_{SU.STO}$	4		0.6		μs
Bus free time between a STOP and START.	t_{BUF}	4.7		1.3		μs

Note: Maximum current is 5mA and load capacitance= 100pF.

7.0 System Control

7.1 System Initialization

7.1.1 Initial Flow

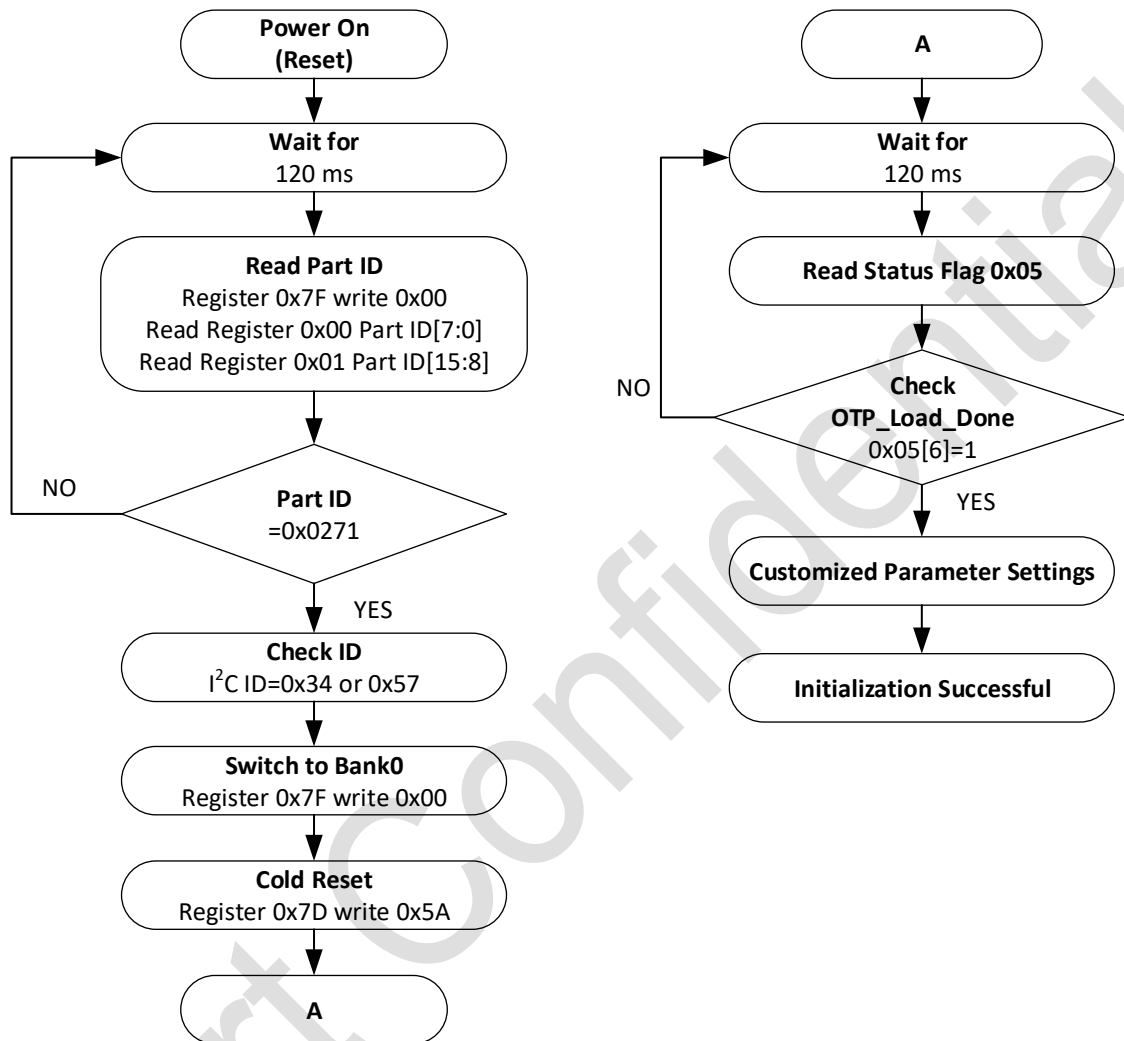


Figure 31. Initialization Flow

Note: If Product ID does not match beyond three times, it is recommended either triggers pin reset or power off/on the sensor again. If the problem persists, do check the I²C interface connection and settings.

7.1.2 Related Register

Usage	Name	Bank	Address
Product ID	PartID[15:0]	0	0x00
		0	0x01
OTP Load Done Flag	OTP_Load_Done [6]	0	0x05

7.2 Registers Access

The host may access register through I²C interface. Refer to [Section 6.0](#) for detail. Any setting will be set in suspend state.

7.2.1 Registers Address Mapping

The sensor has two type of register assignment. They are register bank and register address mapping. Each register bank consists of 7 bits width address mapping (0x00 to 0x7E).

7.2.2 Register Bank Switching

The *Cmd_Bank_Sel* is the register bank location to perform the register bank switching. This bank register address is configured as common address for every register bank called non-bank register.

The register bank location is switching by writing register 0x7F. And switching setting as see as below.

- Write register 0x7F with value 0x00 //switching address mapping register to bank0.
- Write register 0x7F with value 0x01 // switching address mapping register to bank1.

As similar address mapping register is applicable for every register bank, the host need to ensure the correct register bank switching prior perform any register read or write operation.

7.2.3 Related Register

Usage	Name	Bank	Address
Bank Switch	Cmd_Bank_Sel[7:0]	-	0x7F

7.3 System Modes

There are four system modes in operation state: Normal mode, Interrupt mode, Compare mode and One-Shot mode.

In Normal mode, Interrupt mode, Compare mode, the sensor is continuously performing the conversion. These three modes are also known as Burst Mode. The conversion time and report time can be set by register (Refer to [Section 7.6](#) and [Section 7.7](#)).

In One-Shot mode, the sensor performs One-Shot temperature conversion after “sensor enable” command.

The mode selection can be configured through setting the *Alert_Mode* and One-Shot registers in Suspend state. The sensor enters operation after “sensor enable” command as explained in [Section 5.3.3](#). The system mode switching in operation state is prohibited. The sensor is required to transition back to Suspend State followed by entering to the desired mode in the operation state.

Table 18. System Modes Registers

Mode	One-Shot [1]	Alert_Mode[1:0]
Normal	0	0
Interrupt	0	1
Compare	0	2
One-Shot	1	0

Threshold and Hysteresis

■ Threshold

The host needs to configure the upper and lower threshold temperature of Ta and To by setting the Limit Window Register value. The upper and lower threshold are

- Upper threshold= High Limit
- Lower threshold= Lower Limit

The registers are using 11 bits 2's complement to set the threshold values. The resolution is 0.5°C/LSB.

If upper and lower threshold limit is not configured, the compare function will not be enabled.

■ Hysteresis (applicable to Normal and Interrupt mode only)

The hysteresis is used along with threshold. It is designed to avoid the cause of temperature jitters when cross the upper/lower threshold. The host can to configure the hysteresis of Ta and To and registers are using 8 bits to set the hysteresis values. The resolution is 0.5°C/LSB.

If hysteresis register is not configured, the hysteresis function will be disabled.

Table 19. Limit Window and Hysteresis Registers

Bank0	Register	Description
0x6E	Ta_High_Limit[7:0]	Lower byte of High Limit ambient temperature
0x6F	Ta_High_Limit[10:8]	Upper byte of High Limit ambient temperature
0x70	Ta_Low_Limit[7:0]	Lower byte of Low Limit ambient temperature
0x71	Ta_Low_Limit[10:8]	Upper byte of Low Limit ambient temperature
0x72	To_High_Limit[7:0]	Lower byte of High Limit object temperature
0x73	To_High_Limit[10:8]	Upper byte of High Limit object temperature
0x74	To_Low_Limit[7:0]	Lower byte of Low Limit object temperature
0x75	To_Low_Limit[10:8]	Upper byte of Low Limit object temperature
0x76	Ta_Hysteresis[7:0]	Hysteresis of ambient temperature
0x77	To_Hysteresis[7:0]	Hysteresis of object temperature

Bit[10]	Bit[9:0]
Sign	Set Limit value of temperature threshold window

■ Upper and Lower threshold example:

- If threshold value is +30°C (“+” value),

$$\begin{aligned}
 & \blacksquare 30 \div 0.5 = 60 \\
 & \quad = 000\ 0011\ 1100
 \end{aligned}$$

- If threshold value is -30 °C (“-” value),

$$\begin{aligned}
 & \blacksquare 30 \div 0.5 = 60 \\
 & \quad = 000\ 0011\ 1100
 \end{aligned}$$

$$\begin{aligned}
 & \blacksquare 2's\ complement\ conversion &= (111\ 1100\ 0011) + 1 \\
 & &= 111\ 1100\ 0100
 \end{aligned}$$

- Hysteresis example:
 - If hysteresis value is 5°C,
 - $5 \div 0.5 = 10$
 - $= 0000\ 1010$

7.3.1 Normal Mode

In Normal mode, the sensor continuously performs the conversion and deliver alert signals to host. The conversion data is saved to temperature data registers (Bank 0, address 0x06 to 0x0D) and immediately generate alert signals (both alert flag and alert pin are asserted). Once the host read the temperature data, both alert flag and alert pin will be de-asserted. At the same time, the sensor also compare limit against both of upper threshold and lower threshold with hysteresis.

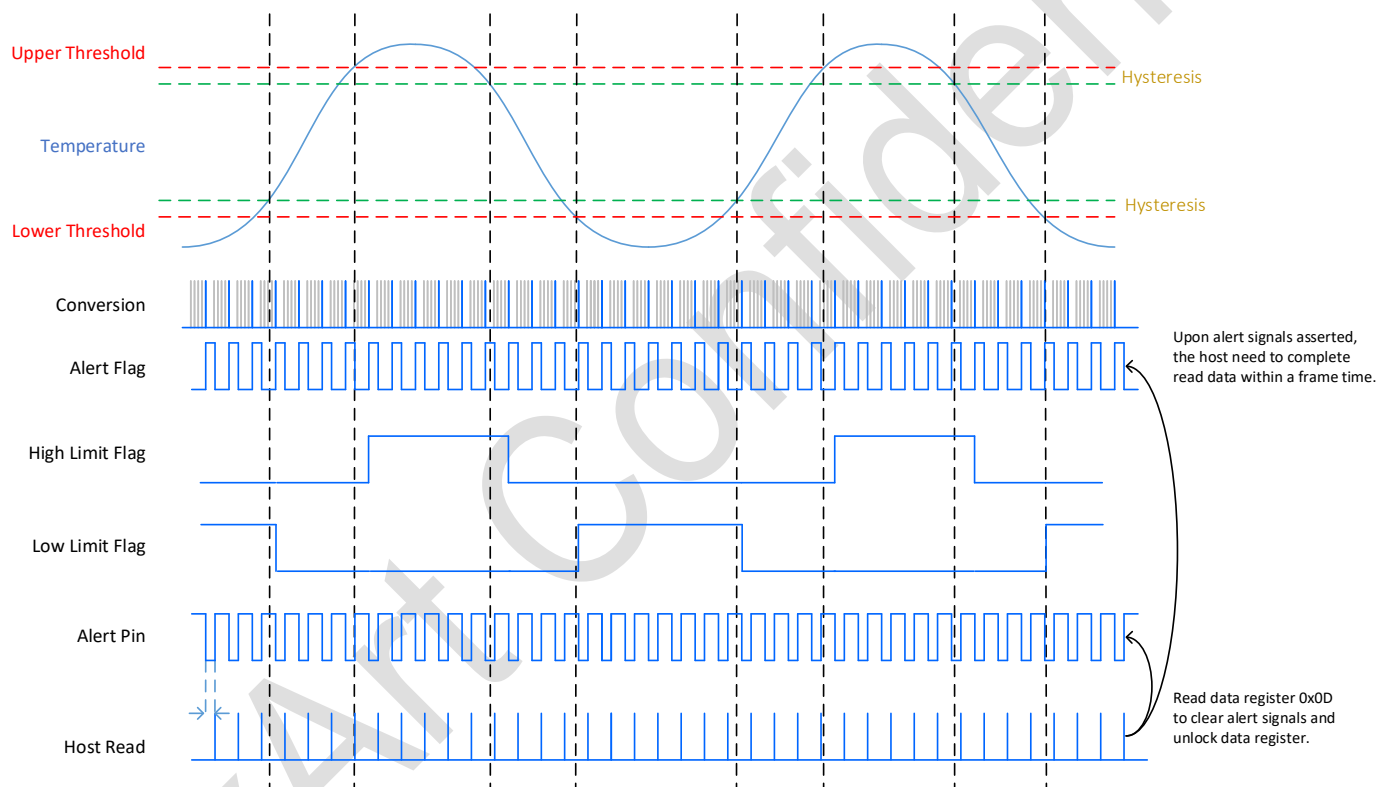


Figure 32. Normal Mode

There are four types of Alert Signals:

- Alert Flag
 1. For each entry to Normal mode, the alert flag is cleared.
 2. The alert flag asserts when the conversion result is available.
 3. The alert flag de-asserts when the host read temperature data register.
 4. Repeats step 2 and 3 until exits this mode.

- High Limit Flag
 1. For each entry to Normal mode, the high limit flag is cleared.
 2. The high limit flag asserts when object temperature rises higher than upper threshold.
 3. The high limit flag de-asserts when object temperature falls lower than “upper threshold limit – hysteresis”.
 4. Repeats step 2 and 3 until exits this mode.

- Low Limit Flag
 1. For each entry to Normal mode, the low limit flag is cleared.
 2. The low limit flag asserts when object temperature falls lower than lower threshold.
 3. The low limit flag de-asserts when object temperature rises higher than “lower threshold limit + hysteresis”.
 4. Repeats step 2 and 3 until exits this mode.

- Alert Pin (active low)
 1. For each entry to Normal mode, the alert pin is reset to high state.
 2. The alert pin asserts when the conversion result is available.
 3. The alert pin de-asserts when host read temperature data register.
 4. Repeats step 2 and 3 until exits this mode.

Table 20. Normal Mode Alert Signals

Alert Signals	Initial State	Temperature rise above upper threshold	Temperature fall below “upper threshold - hysteresis”	Temperature fall below lower threshold	Temperature rise above “lower threshold + hysteresis”
Alert Flag	0	Assert	Assert	Assert	Assert
High Limit Flag	0	Assert	De-assert	No impact	No impact
Low Limit Flag	0	No impact	No impact	Assert	De-assert
Alert Pin	High	Assert; De-assert after host read	Assert; De-assert after host read	Assert; De-assert after host read	Assert; De-assert after host read

7.3.2 Interrupt Mode

In Interrupt mode, the host needs to configure the upper and lower threshold temperature register first. The sensor measures temperature and saves it to temperature data registers (Bank 0, address 0x06 to 0x0D) periodically. The sensor delivers alert pin to host during temperature rise above upper threshold or fall below lower threshold. This mode is used for monitoring temperature when object temperature rises above upper threshold or falls below lower threshold.

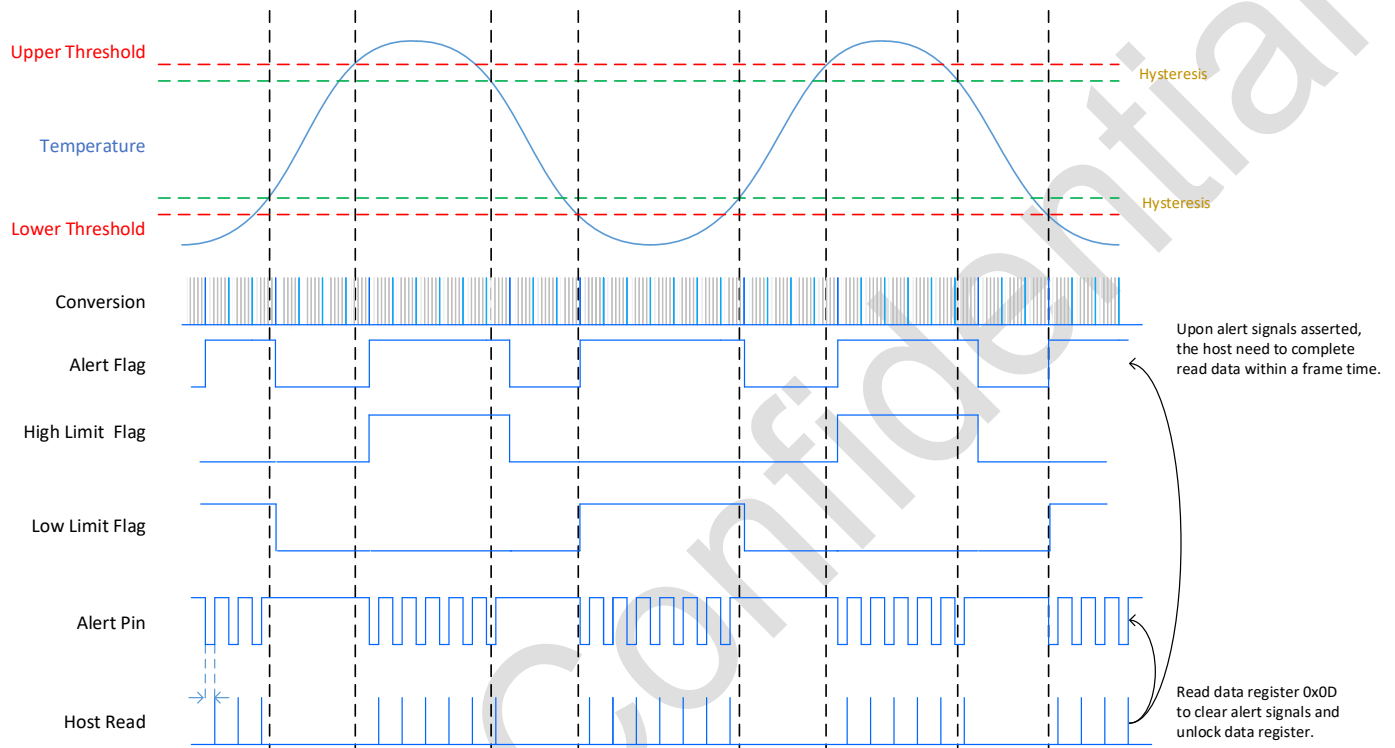


Figure 33. Interrupt Mode

There are four types of Alert Signals:

- Alert Flag
 1. For each entry to Interrupt mode, the alert flag is cleared.
 2. The alert flag asserts when the conversion result is available.
 3. The alert flag de-asserts when host read temperature data register.
 4. Repeats step 2 and 3 until exits this mode.
- High Limit Flag
 1. For each entry to Interrupt mode, the high limit flag is cleared.
 2. The high limit flag asserts when object temperature rises higher than upper threshold.
 3. The high limit flag de-asserts when object temperature falls lower than “upper threshold limit – hysteresis”.
 4. Repeats step 2 and 3 until exits this mode.

- Low Limit Flag
 1. For each entry to interrupt mode, the low limit flag is cleared.
 2. The low limit flag asserts when object temperature falls lower than lower threshold.
 3. The low limit flag de-asserts when object temperature rises higher than “lower threshold limit + hysteresis”.
 4. Repeats step 2 and 3 until exits this mode.

- Alert Pin (active low)
 1. For each entry to Interrupt mode, the alert pin is reset to high state.
 2. The alert pin asserts when the object temperature rises above upper threshold or falls below lower threshold.
 3. The alert pin de-asserts when host read temperature data register.
 4. Repeats step 2 and 3 until exits this mode.

Table 21. Interrupt Mode Alert Signals

Alert Signals	Initial State	Temperature rise above upper threshold	Temperature fall below “upper threshold - hysteresis”	Temperature fall below lower threshold	Temperature rise above “lower threshold + hysteresis”
Alert Flag	0	Assert	De-assert	Assert	De-assert
High Limit Flag	0	Assert	De-assert	No impact	No impact
Low Limit Flag	0	No impact	No impact	Assert	De-assert
Alert Pin	High	Assert; De-assert after host read	No impact	Assert; De-assert after host read	No impact

7.3.3 Compare Mode

In Compare mode, the host needs to configure the upper and lower threshold temperature registers first. The conversion function is on-going at any time during the conversion state. When threshold criteria are met with alert flag changing, the conversion data is saved to temperature data registers (Bank 0, address 0x06 to 0x0D) and the alert signals activated. The benefit of this mode is to monitor and compare conversion result automatically without additional host resource.

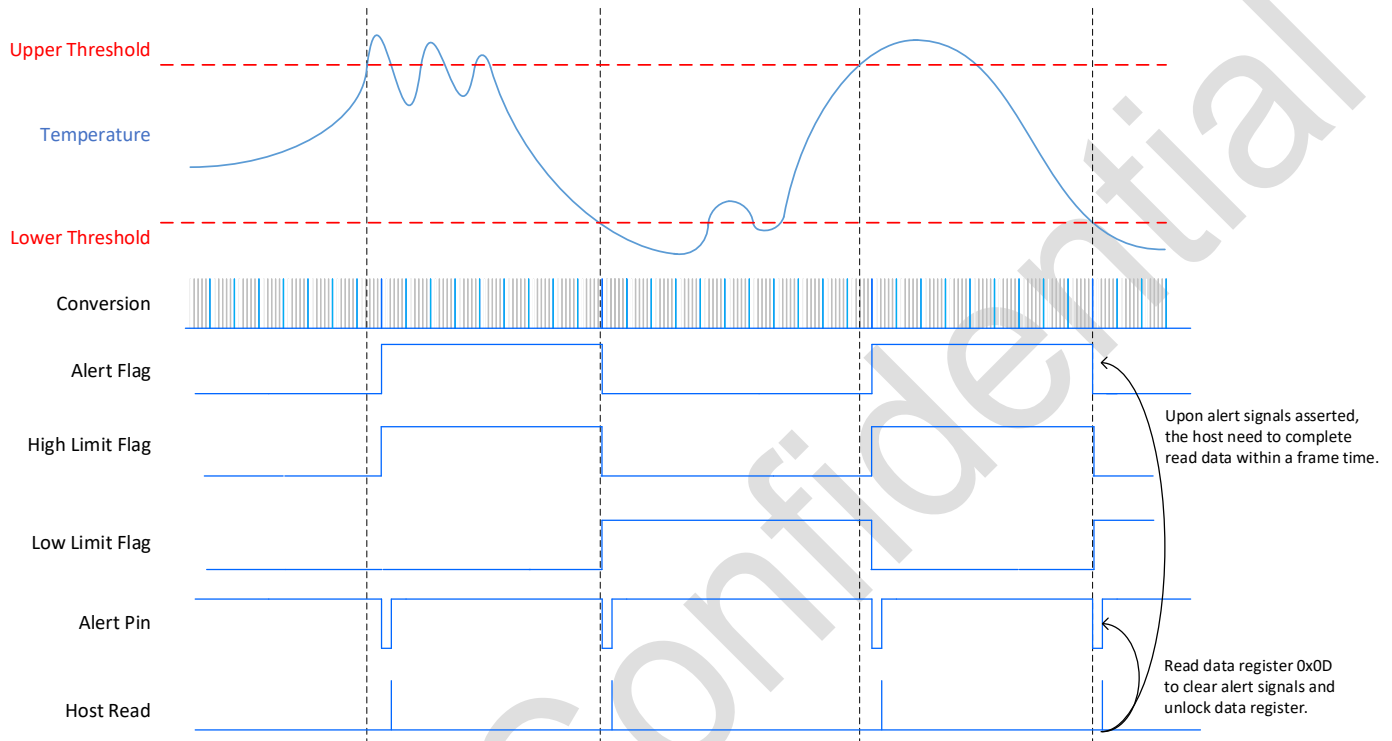


Figure 34. Compare Mode

There are four types of Alert Signals:

- Alert Flag
 1. For each entry to Compare mode, the alert flag is cleared.
 2. The alert flag will be set and hold when
 - a. First conversion result rise above upper threshold before hit lower threshold.
 - b. First conversion result fall below lower threshold before hit upper threshold.
 3. The alert flag will be cleared and hold when
 - a. Continuous from 2a, first conversion result fall below lower threshold before hit upper threshold again.
 - b. Continuous from 2b, first conversion result rise above upper threshold before hit lower threshold again.
 4. Repeats step 2 and 3 until exits this mode.
- High Limit Flag
 1. For each entry to Compare mode, the high limit flag is cleared.
 2. The high limit flag will be asserted when first conversion result rise above upper threshold before hit lower threshold.
 3. The high limit flag de-asserts when-first conversion result fall below lower threshold before hit upper threshold.
 4. Repeats step 2 and 3 until exits this mode.

- Low Limit Flag
 1. For each entry to Compare mode, the low limit flag is cleared.
 2. The low limit flag will be asserted when first conversion result fall below lower threshold before hit upper threshold.
 3. The lower limit flag de-asserts when first conversion result rise above upper threshold before hit lower threshold again.
 4. Repeats step 2 and 3 until exits this mode.

- Alert Pin (active low)
 1. For each entry to Compare mode, the alert pin is reset at high state.
 2. The alert pin will be asserted when alert flag change.
 3. The alert pin de-assert when host read temperature data register.
 4. Repeats step 2 and 3 until exits this mode.

Table 22. Compare Mode Alert Signals

Alert Signals	Initial State	First one temperature rises above upper threshold	Temperature fall below upper threshold	First one temperature falls below lower threshold	Temperature rise above lower threshold
Alert Flag	0	Assert	No impact	De-assert	No impact
High Limit Flag	0	Assert	No impact	De-assert	No impact
Low Limit Flag	0	De-assert	No impact	Assert	No impact
Alert Pin	High	Assert; De-assert after host read	No impact	Assert; De-assert after host read	No impact

7.3.4 One-Shot Mode

In One-Shot mode, host trigger a single temperature conversion by command. The sensor performs the conversion once and compare upper threshold and lower threshold. The conversion data is saved to temperature data registers (Bank 0, address 0x06 to 0x0D) and immediately generate alert signals. The sensor returns to the suspend state automatically upon complete of One-Shot operation. This mode is used for reducing power consumption when continuous monitoring is not required.

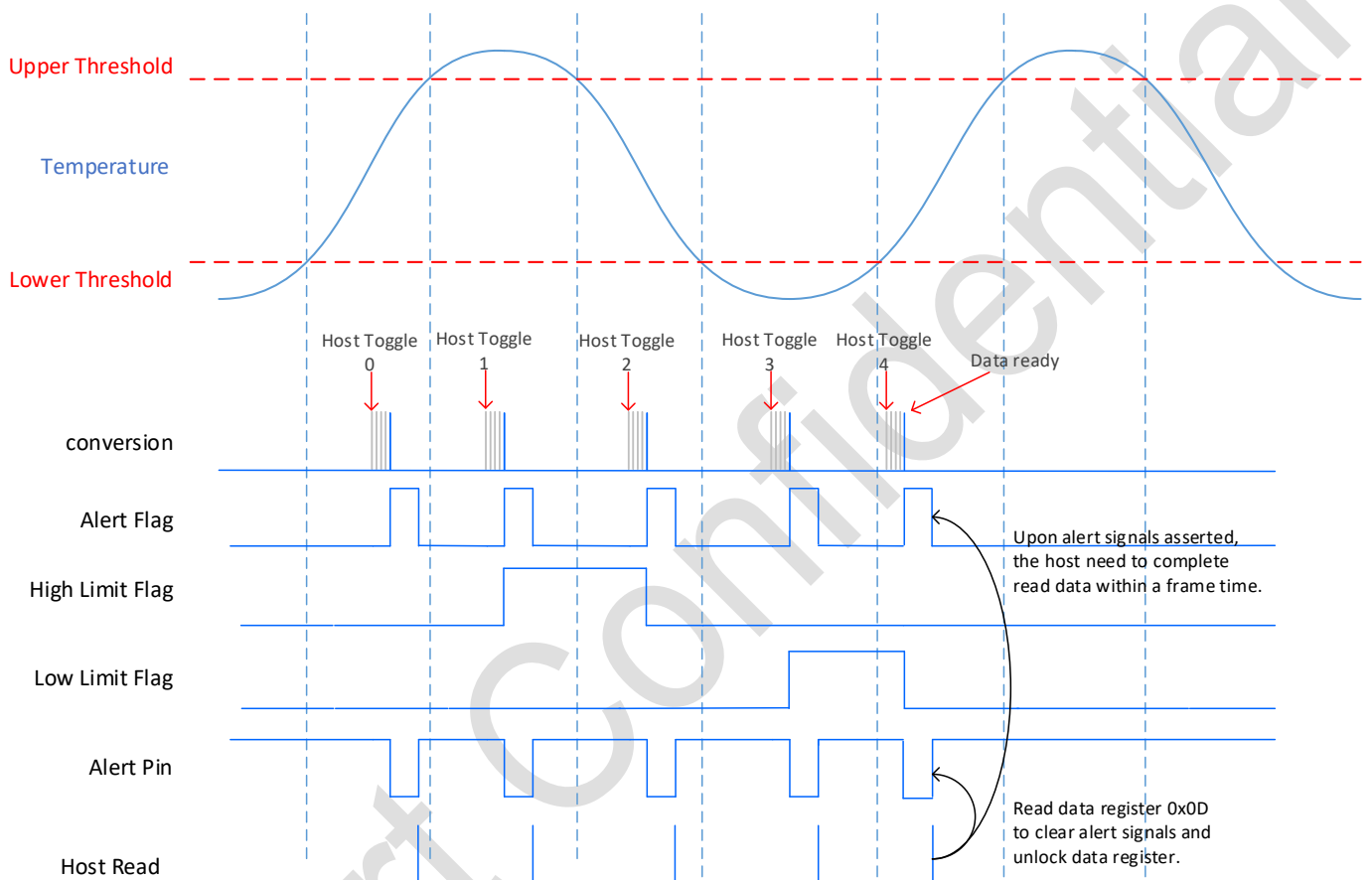


Figure 35. One-Shot Mode

There are four types of Alert Signals:

- Alert Flag
 1. For each entry to One-Shot mode, the alert flag is cleared.
 2. The alert flag asserts when the conversion result is available.
 3. The alert flag de-asserts when host read temperature data register.
- High Limit Flag
 1. For each entry to One-Shot mode, the high limit flag is cleared.
 2. The high limit flag will be set when conversion result rise above upper threshold.

- Low Limit Flag
 1. For each entry to One-Shot mode, the low limit flag is cleared.
 2. The low limit flag will be set when conversion result fall below lower threshold.
- Alert Pin (active low)
 1. For each entry to One-Shot mode, the alert pin is reset at high state.
 2. Alert pin will be asserted when the conversion result is available.
 3. The alert pin will be de-asserted when host accessed temperature data registers.

Table 23. One-Shot Mode Alert Signals

Alert Signals	Initial State	(Host toggle 1) Temperature Rise Above Upper Threshold	(Host toggle 2) Temperature Fall Below Upper Threshold	(Host toggle 3) Temperature Fall Below Lower Threshold	(Host toggle 4) Temperature Rise Above Lower Threshold
Alert Flag	0	Assert; De-assert after host read	Assert; De-assert after host read	Assert; De-assert after host read	Assert; De-assert after host read
High Limit Flag	0	Assert	De-assert	No impact	No impact
Low Limit Flag	0	No impact	No impact	Assert	De-assert
Alert Pin	High	Assert	Assert	Assert	Assert

7.3.5 Related Register

Usage	Name	Bank	Address	Bit Field
System Mode	Alert_Mode[1:0]	0x00	0x03	bit [1:0]
	One-Shot		0x26	bit [1]
Limit Window	Ta_High_Limit[7:0]	0x00	0x6E	bit [7:0]
	Ta_High_Limit[10:8]		0x6F	bit [10:8]
	Ta_Low_Limit[7:0]		0x70	bit [7:0]
	Ta_Low_Limit[10:8]		0x71	bit [10:8]
	To_High_Limit[7:0]		0x72	bit [7:0]
	To_High_Limit[10:8]		0x73	bit [10:8]
	To_Low_Limit[7:0]		0x74	bit [7:0]
	To_Low_Limit[10:8]		0x75	bit [10:8]
Hysteresis	Ta_Hysteresis[7:0]	0x00	0x76	bit [7:0]
	To_Hysteresis[7:0]		0x77	bit [7:0]
Status Register	To_Low_Limit_Flag	0x00	0x05	bit [4]
	To_High_Limit_Flag			bit [3]
	Ta_Low_Limit_Flag			bit [2]
	Ta_High_Limit_Flag			bit [1]
	Alert_Flag			bit [0]

7.4 Output Access

Below is a reference of the regular data transmission for comparison purpose. The sensor is continuously performing the conversion in conversion state. The conversion data is saved to data registers (Bank 0, 0x06 to 0x0D) until next data has conversion ready to replace the previous data. Then, the sensor sends the alert signals to host. When the host reads the data registers (Bank0, 0x06 to 0x0C), the data registers remain unchanged until the host triggers to de-assert the alert signals.

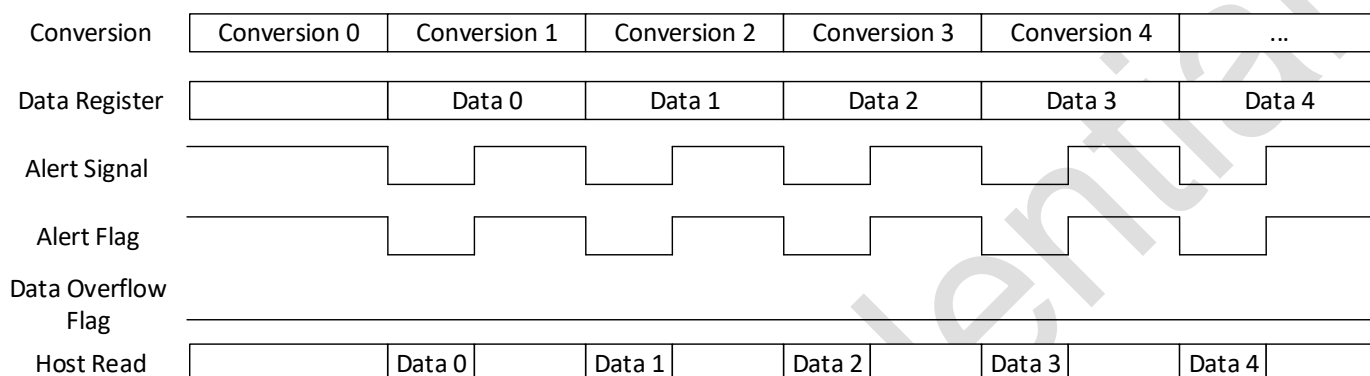


Figure 36. Regular Data Transmission

The host can use either Interrupt or polling method to access sensor output.

7.4.1 Alert Signals

The sensor outputs the alert signals to access the temperature data and the status flags. The alert signals assertion/de-assertion as below.

- Alert signals assertion

When the data conversion is completed, the alert signals will be generated to notify the host for assessing the data.

- Alert signals de-assertion

There are two methods to de-assert the alerts signals.

- Method 1: Accessing the temperature data register (Bank0, 0x0D) with *Host_Clear_Enable*= 0
After reading the temperature data register (Bank0, 0x0D), the alert signals will be de-asserted automatically.
- Method 2: Manual request with *Host_Clear_Enable*= 1

The alert signals can be cleared manually based on demand as follow.

Write register 0x03 bit[5] with value 1 (*Host_Clear_Enable*) to enable the manual request function.

Write register 0x05 bit[5] with value 1 (*Host_Clear_Interrupt*) to execute “the host clear” function as needed.

7.4.2 Interrupt Method

The alert pin is act as indicator if data is ready for the host to read. If alert pin is asserted, the data registers are pending the host to read. Otherwise, the alert pin is de-asserted when the register is read.

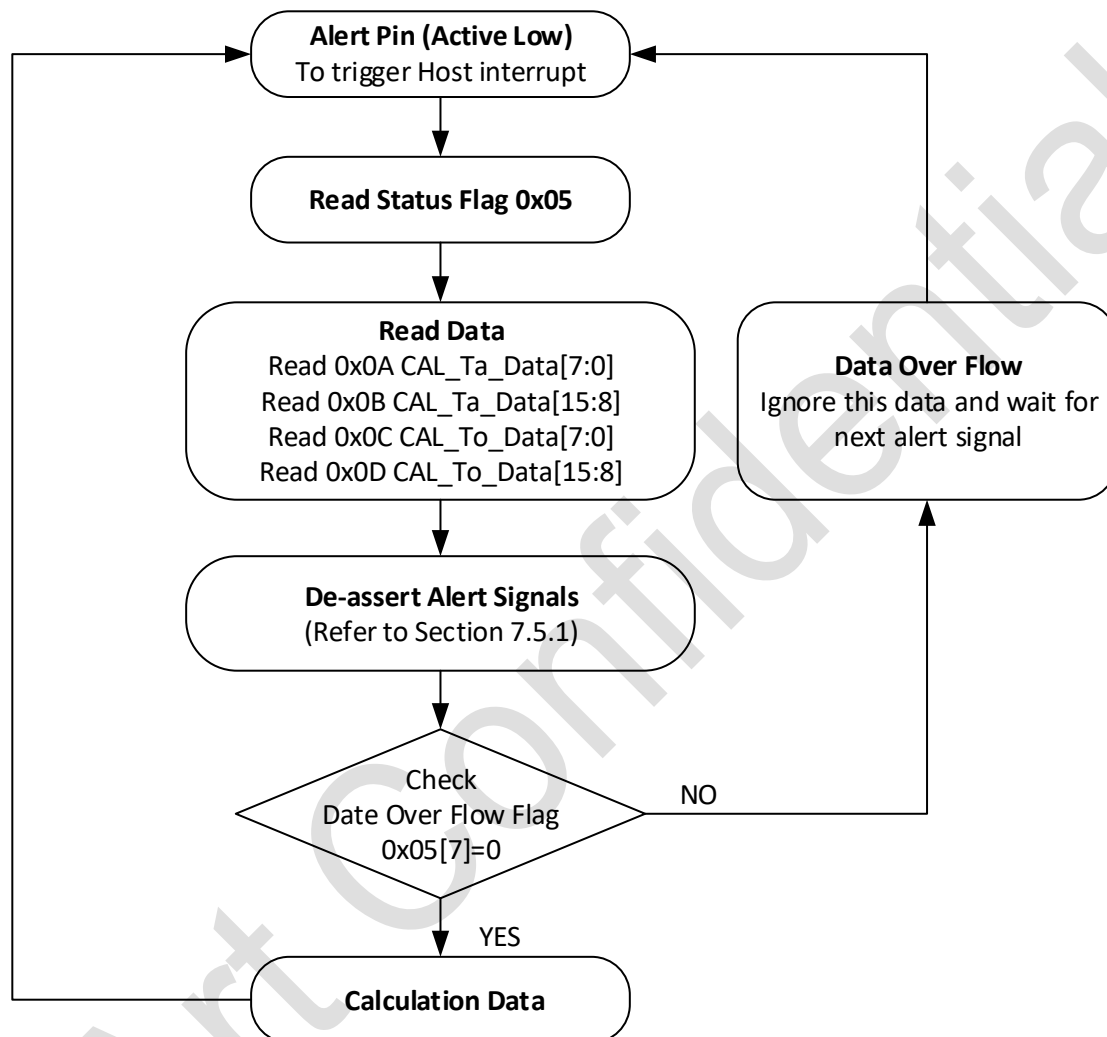


Figure 37. Interrupt Method

7.4.3 Polling Method

The polling method applicable only for Normal mode. By reading and checking register *Status* (Bank 0, address 0x05) periodically, the host can get the temperature data through I²C interface. The delay time may be varying where it depends on the host capability and requirement of applications.

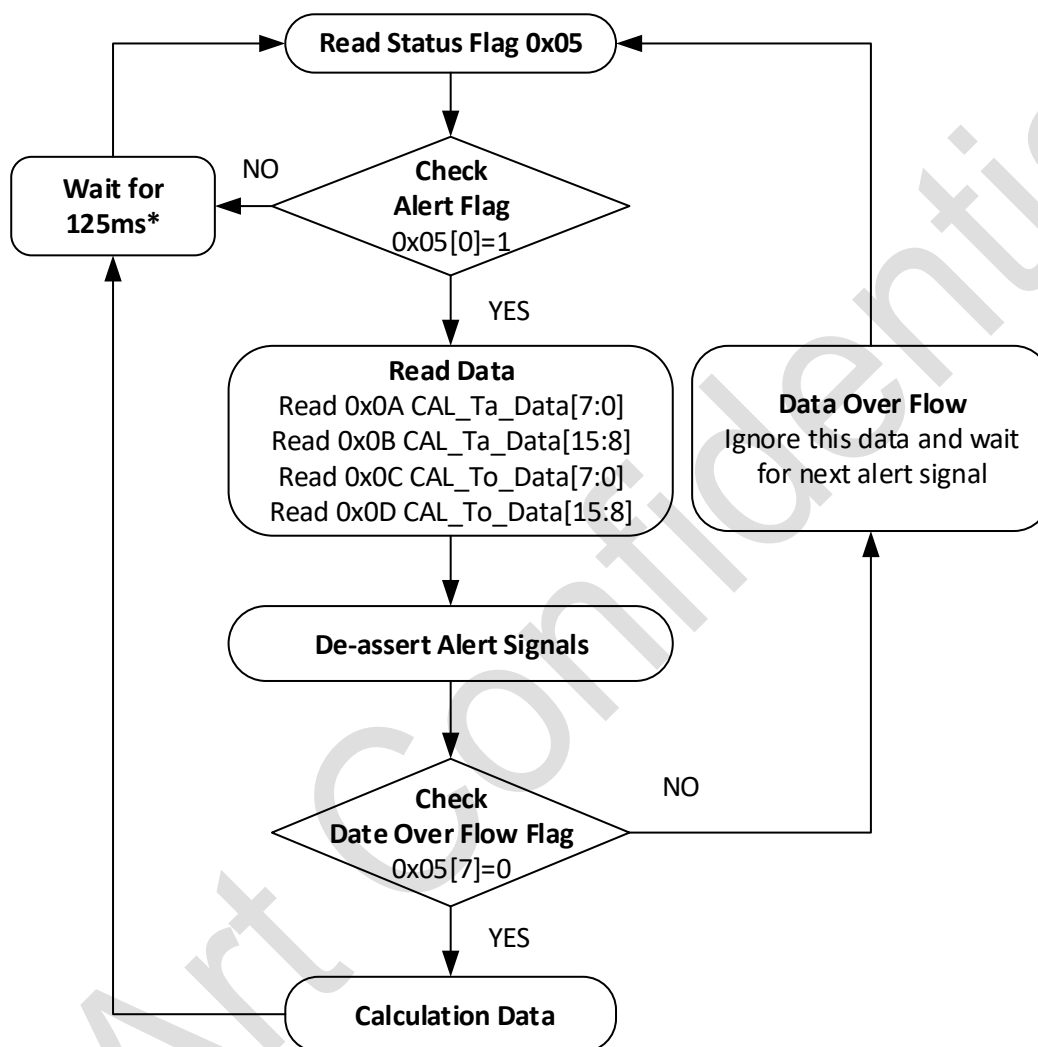


Figure 38. Polling Method

Note: Waiting time setting is based on report rate setting. Waiting time $\leq (\text{report time}) \div 2$

7.4.4 Data Loss and Corruption

The possible data loss (1) is when the conversion data 2 is awaiting the alert signals to be de-asserted. As the data 1 in data registers is not read completely causes the registers in lock state and the conversion data 2 is unable to save to data registers. Therefore, it causes data 2 loss and data overflow flag is set.

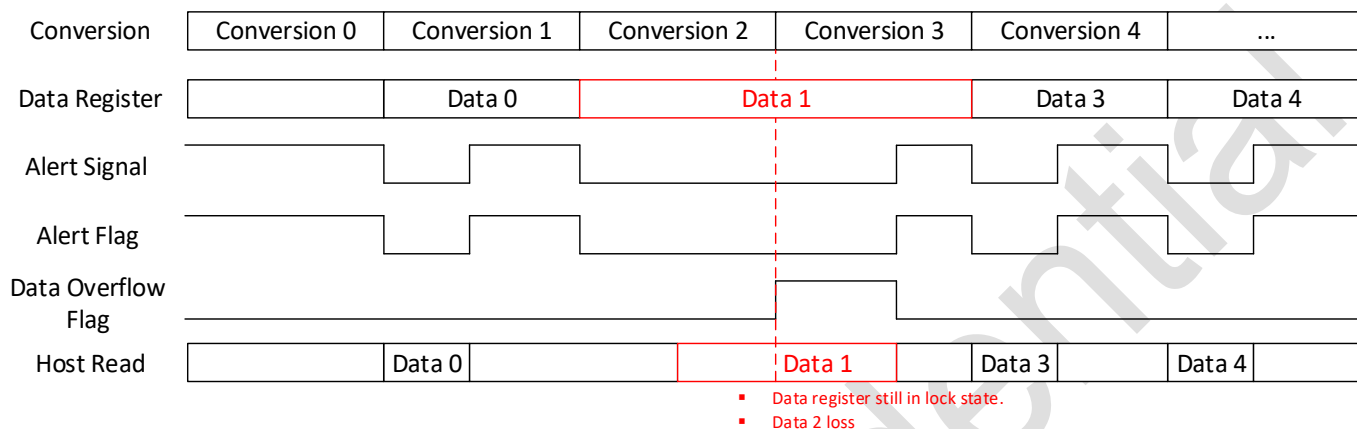


Figure 39. Possible Data Loss (1)

The data is continuously saved and keep to data register (Bank 0, 0x06 to 0x0D) until next data has conversion ready to replace the previous data. The possible data loss (2) is when the host begins to read data behind the conversion 1, the host will read the latest data 1 since the data 0 cleared. Therefore, it causes to data 0 loss.

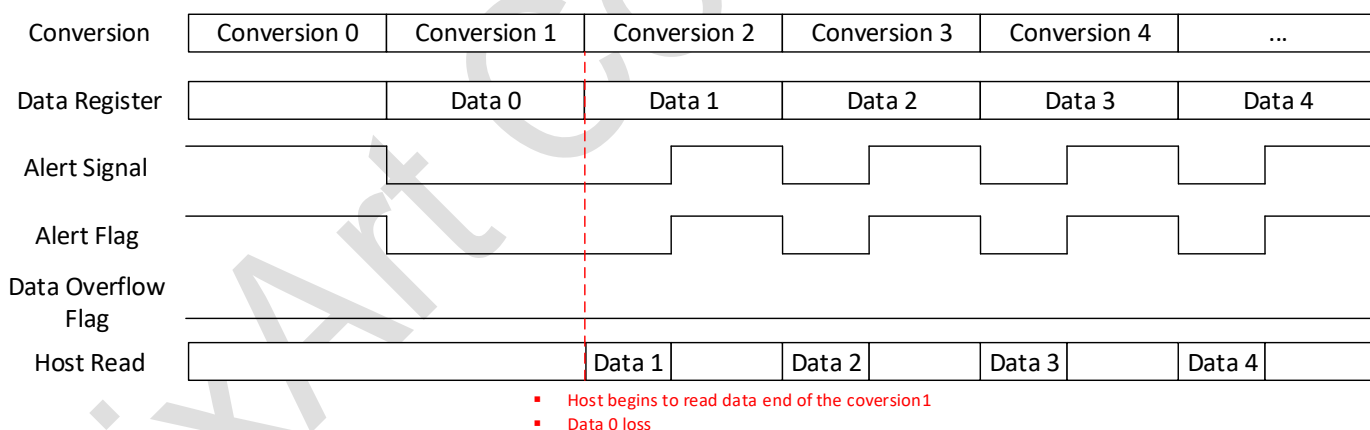


Figure 40. Possible Data Loss (2)

Possible options to overcome the data loss:

- Prolong the report rate
- Improve the host data processing time

7.4.5 Related Register

Usage	Name	Bank	Address	Bit Field
Host Clear Enable	Host_Clear_Enable	0x00	0x03	bit [5]
Host Clear Interrupt	Host_Clear_Interrupt	0x00	0x05	bit [5]
Data Overflow	Data_Overflow_Flag	0x00	0x05	bit [7]

7.5 Output

The sensor can output temperature data and indicator status. The temperature data for Ta and To be stored in data registers (Bank 0; address 0x06 to 0x0D). The sensor can provide two types of conversion result. They are ADC raw data and temperature (computed and calibrated with built-in algorithm).

Table 24. Temperature Data Registers

Address (Bank0)	Register Bit Field	Description
0x06	DSP_Ta_Data[7:0]	Lower byte of Ta ADC raw data
0x07	DSP_Ta_Data[15:8]	Upper byte of Ta ADC raw data
0x08	DSP_To_Data[7:0]	Lower byte of To ADC raw data
0x09	DSP_To_Data[15:8]	Upper byte of To ADC raw data
0x0A	CAL_Ta_Data[7:0]	Lower byte of Ta ambient temperature
0x0B	CAL_Ta_Data[15:8]	Upper byte of Ta ambient temperature
0x0C	CAL_To_Data[7:0]	Lower byte of To object temperature
0x0D	CAL_To_Data[15:8]	Upper byte of To object temperature

7.5.1 ADC Raw Data Registers

The Ambient ADC (*DSP_Ta_Data*) and Object ADC (*DSP_To_Data*) data format is using 16 bits 2's complement.

Bit[15]	Bit[14:0]
Sign bit	Ambient ADC Code= <i>DSP_Ta_Data</i> [14:0] Object ADC Code= <i>DSP_To_Data</i> [14:0]

For example, read the either *DSP_Ta_Data*[15:0] or *DSP_To_Data*[15:0] to obtain the raw data and check for the sign bit.

- If the Bit[15:0] = 0110 0011 1011 1111, then

Sign bit= 0

$$\begin{aligned} DSP_Ta/To_Data [15:0] &= 0110 0011 1011 1111 \\ &= 25535 \end{aligned}$$

- If the Bit[15:0] = 1001 1100 0100 0001,

Sign bit= 1

$$DSP_Ta/To_Data [15:0] - 1 = 1001 1100 0100 0000$$

$$\begin{aligned} \text{Bits inverted} &= 0110 0011 1011 1111 \\ &= 25535 \end{aligned}$$

Adding a negative sign to have effective value of *DSP_Ta/To_Data* = -25535

7.5.2 Temperature Data Registers

The ambient temperature (CAL_Ta_Data) and Object temperature (CAL_To_Data) data format is using 16 bits 2's complement. The output scale is 0.03125°C/LSB.

Bit[15]	Bit[14:0]
Sign bit	Ambient temperature= $CAL_Ta_Data[14:0]$ Object temperature= $CAL_To_Data[14:0]$

For example, read the either $CAL_Ta_Data[15:0]$ or $CAL_To_Data[15:0]$ to obtain the raw data and check for the sign bit.

- If the Bit[15:0] = 0000 0011 1100 0000, then

Sign bit = 0

$$\begin{aligned} CAL_Ta/To_Data[15:0] &= 0000\ 0011\ 1100\ 0000 \\ &= 960 \end{aligned}$$

$$\begin{aligned} \text{Effective Temperature} &= 960 \times 0.03125 \\ &= 30^{\circ}\text{C} \end{aligned}$$

- If the Bit[15:0] = 1111 1100 0100 0000, then

Sign bit = 1

$$\begin{aligned} CAL_Ta/To_Data[15:0] - 1 &= 1111\ 1100\ 0011\ 1111 \\ \text{Bits inverted} &= 0000\ 0011\ 1100\ 0000 \\ &= 960 \end{aligned}$$

Adding a negative sign to have effective value of $CAL_Ta/To_Data = -960$

$$\begin{aligned} \text{Effective Temperature} &= -960 \times 0.03125 \\ &= -30^{\circ}\text{C} \end{aligned}$$

7.5.3 Related Register

Usage	Name	Bank	Address	Bit Field
Ta ADC Raw Data	DSP_Ta_Data[7:0]	0x00	0x06	bit [7:0]
	DSP_Ta_Data[15:8]		0x07	bit [15:8]
To ADC Raw Data	DSP_To_Data[7:0]	0x00	0x08	bit [7:0]
	DSP_To_Data[15:8]		0x09	bit [15:8]
Ta Ambient Temperature	CAL_Ta_Data[7:0]	0x00	0x0A	bit [7:0]
	CAL_Ta_Data[15:8]		0x0B	bit [15:8]
To Object Temperature	CAL_To_Data[7:0]	0x00	0x0C	bit [7:0]
	CAL_To_Data[15:8]		0x0D	bit [15:8]

7.6 Conversion

Each conversion data is an average of multiple ADC samples to reduce the noise effect. The host can configure the sampling numbers through *Burst_Num_Sel* register (Bank0, 0x20) for Ta (M) and To (N). The sampling number is based on formula of 2^M and 2^N . The sensor conversion default setting is 2^9 for Ta and 2^{12} for To.

The conversion time is combination of the Ta and To sampling time. The conversion time needs to be less than Report time (Frame Time).

Table 25. Conversion Time

Ta (M)	To (N)	Conversion Time (ms)
8	10	68
9	10	81
10	10	107
8	11	119
9	11	132
10	11	158
8	12	222
9	12	235(default)
10	12	260
10	13	463
10	14	875

For example,

Ta samples (M)= 9, the Ta data is averaging 512 (2^9) ADC samples

To samples (N)= 12, the To data is averaging 4096 (2^{12}) ADC samples

The conversion time is 235ms for one conversion

7.6.1 Related Register

Usage	Name	Bank	Address	Bit Field
Sampling Number	To_Burst_Num_Sel[3:0]	0x00	0x20	bit [3:0]
	Ta_Burst_Num_Sel[3:0]			bit [7:4]

7.7 Report Rate

The host configures the Report Rate through *Report Rate* register (Bank 0, 0x21 to 0x23).

$$\text{Frame time(sec)} = (1 \div 20K) \times \text{Burst_Freq_Sel} [19:0] \times 16$$

$$\text{Report Rate (Hz)} = 1 \div \text{Frame time}$$

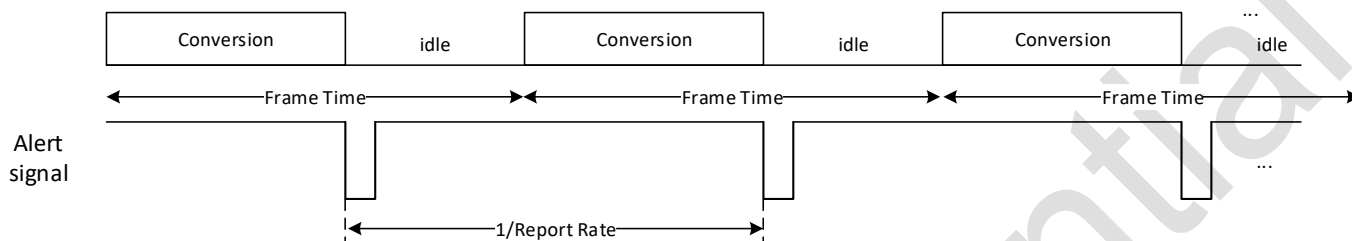


Figure 41. Report Rate

For example,

Report Rate = 1Hz.

$$0x21 (\text{Burst_Freq_Sel} [7:0]) = 0xE2$$

$$0x22 (\text{Burst_Freq_Sel} [15:8]) = 0x04$$

$$0x23 (\text{Burst_Freq_Sel} [19:16]) = 0x00$$

$$\begin{aligned} \text{Report Rate} &= 1 \div [(1 \div 20K) \times 1250 \times 16] \\ &= 1 \text{ Hz} \end{aligned}$$

Note: The Conversion time must within the report period.

7.7.1 Related Register

Usage	Name	Bank	Address	Bit Field
Report Rate Setting	Burst_Freq_Sel[7:0]	0x00	0x21	bit [7:0]
	Burst_Freq_Sel[15:8]		0x22	bit [7:0]
	Burst_Freq_Sel[19:16]		0x23	bit [3:0]

7.8 Filter

The sensor provides three types of digital filter. They are IIR Filter (Infinite Impulse Response filter), Normal Average Filter and Moving Average Filter that can be easily customized for virtually any temperature data average desired by the customer application by changing the content of *Filter_Mode* registers. However, digital filter may impact the response time of the sensor.

Table 26. Filter Mode Registers

Bank1	Register	Description
0x5B[4:3]	<i>Filter_Mode</i> [1:0]	0: IIR Mode 1: Moving Average Mode 2: Normal Average Mode

7.8.1 IIR Filter

The sensor is integrated IIR digital filter to improve noise performance of the thermal sensing output. This is a smooth temperature output and is used for the detection application. However, it may impact the response time of the sensor.

The IIR filter can configure IIR level through *IIR level setting* register as below.

Table 27. IIR Level

IIR Level	Bank1 0x5B[2:0]	n-1	n
0 (Default)	000	0	1
1	001	0.125	0.875
2	010	0.25	0.75
3	011	0.375	0.625
4	100	0.5	0.5
5	101	0.625	0.375
6	110	0.75	0.25
7	111	0.875	0.125

7.8.2 Normal Average Filter

The Sensor supports normal average filter. It may improve the SNR under low report rate condition.

- Frame average number: 1, 2, 4, 8

For example, setting frame average number= 4

Data1= conversion 1/2/3/4 data average

Data2= conversion 5/6/7/8 data average

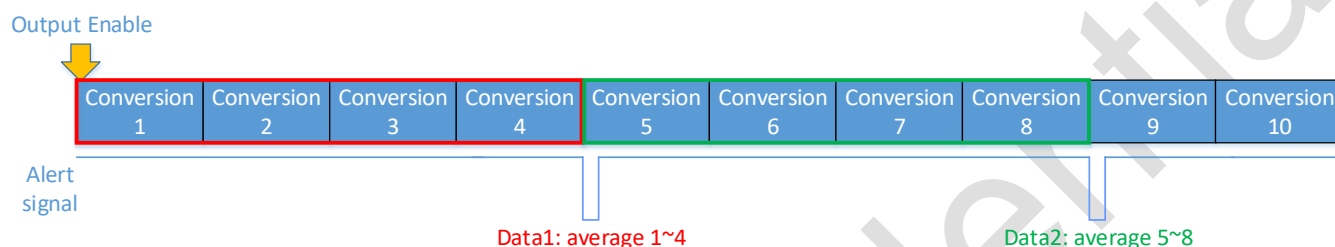


Figure 42. Normal Average

The normal average level can configure frame average number through *FIR_Stage[1:0]* register as below.

Table 28. Frame Average Level

Frame Average Level	Bank1 0x5B[6:5] <i>FIR_Stage[1:0]</i>	Frame Average Number
0	00	1
1	01	2
2	10	4
3	11	8

7.8.3 Moving Average Filter

The Sensor supports Moving Average filter. It may improve the SNR and smoothen the temperature data output under high report rate condition. The moving average levelling can be configured the frame average number through *FIR_Stage[1:0]* register. Refer to Table 28. Frame Average Levelling for frame average number: 1, 2, 4 and 8.

For example, setting frame average number= 4

- New Data1= conversion 1 data
- New Data2= conversion 1/2 data average
- New Data3= conversion 1/2/3 data average
- New Data4= conversion 1/2/3/4 data average
- New Data5= conversion 2/3/4/5 data average

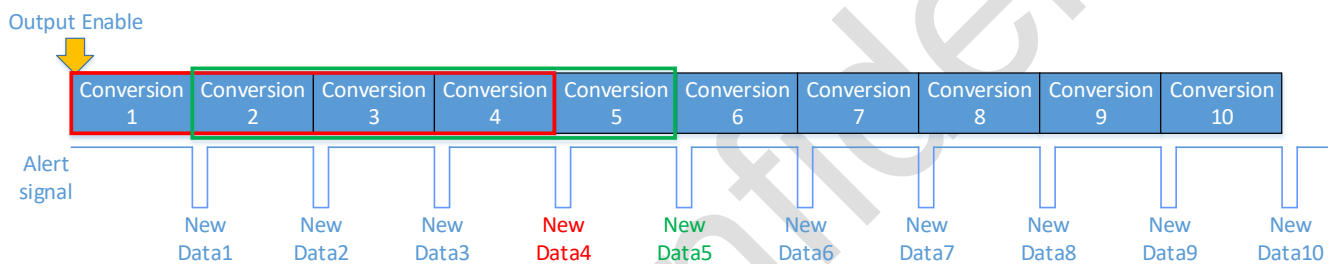


Figure 43. Moving Average Filter

7.8.4 Related Register

Usage	Name	Bank	Address	Bit Field
Filter Setting	IIR level[2:0]	0x01	0x5B	bit [2:0]
	Filter_Mode[1:0]			bit [4:3]
	FIR_Stage[1:0]			bit [6:5]

7.9 Emissivity

Emissivity is the estimate of an object's ability to emit infrared energy. Emitted energy indicates the object temperature. The sensor has adjustable emissivity correction to ensure temperature accuracy when measuring other materials with a low emissivity such as shiny metals or Aluminum. The sensor supports programmable object emissivity that can be customized by the customer for any emissivity coefficient range from 0 to 1 with 100 steps. The emissivity scale is 0.01 LSB.

7.9.1 Related Register

Usage	Name	Bank	Address	Bit Field
Emissivity Setting	<i>Emissivity[6:0]</i>	0x01	0x50	bit [6:0]

7.10 Decay

As the cover's decay effect impacts the sensor sensitivity and T_o accuracy, it needs to be compensated through configuring the decay setting register. Decay range from 0% to 99.5% with 0.5% steps.

For example, Decay % = 32%

$$\begin{aligned} 32 \div 0.5 &= 64 \\ &= 0100\ 0000 \\ &= 0x40 \end{aligned}$$

7.10.1 Related Register

Usage	Name	Bank	Address	Bit Field
Decay Setting	Decay[7:0]	0x01	0x51	bit [7:0]

7.11 Memory

The sensor has integrated memory to save the non-disclosure calibration and thermal transient coefficients' parameters for calibration and computing function to output the T_a and T_o temperature data.

8.0 Register

8.1 Register Map

There are 2 banks register for accessing the firmware setting to customize for difference usage and applications.

Table 29. Register Bank 0

Address	Bit Field	Name	Access	Bit Field Default Value	Register Default Value
0x00		PartID[7:0]	R		0x71
0x01		PartID[15:8]	R		0x02
0x02		Version ID	R		0x00
0x03	bit[5]	Host_Clear_Enable	R/W	0	0x00
	bit[1:0]	Alert_Mode[1:0]	R/W	0	
0x04		Output_Enable[7:0]	R/W		0x00
0x05	bit[7]	Data_Overflow_Flag	R	0	0x00
	bit[6]	OTP_Load_Done	R	0	
	bit[5]	Host_Clear_Interrupt	R/W	0	
	bit[4]	To_Low_Limit_Flag	R	0	
	bit[3]	To_High_Limit_Flag	R	0	
	bit[2]	Ta_Low_Limit_Flag	R	0	
	bit[1]	Ta_High_Limit_Flag	R	0	
	bit[0]	Alert_Flag	R	0	
0x06		DSP_Ta_Data[7:0]	R		0x00
0x07		DSP_Ta_Data[15:8]	R		0x00
0x08		DSP_To_Data[7:0]	R		0x00
0x09		DSP_To_Data[15:8]	R		0x00
0x0A		CAL_Ta_Data[7:0]	R		0x00
0x0B		CAL_Ta_Data[15:8]	R		0x00
0x0C		CAL_To_Data[7:0]	R		0x00
0x0D		CAL_To_Data[15:8]	R		0x00
0x20	bit[7:4]	Ta_Burst_Num_Sel[3:0]	R/W	9	0x9C
	bit[3:0]	To_Burst_Num_Sel[3:0]	R/W	12	
0x21		Burst_Freq_Sel[7:0]	R/W		0x38
0x22		Burst_Freq_Sel[15:8]	R/W		0x01
0x23	bit[3:0]	Burst_Freq_Sel[19:16]	R/W	0	0x00
0x26	bit[1]	One-Shot	R/W	0	0x00
0x2F	bit[0]	ALERT_Open_Drain	R/W	0	0x00
0x6E		Ta_High_Limit[7:0]	R/W		0xFF
0x6F	bit[2:0]	Ta_High_Limit[10:8]	R/W	3	0x03
0x70		Ta_Low_Limit[7:0]	R/W		0x01
0x71	bit[2:0]	Ta_Low_Limit[10:8]	R/W	4	0x04
0x72		To_High_Limit[7:0]	R/W		0xFF

Address	Bit Field	Name	Access	Bit Field Default Value	Register Default Value
0x73	bit[2:0]	To_High_Limit[10:8]	R/W	3	0x03
0x74		To_Low_Limit[7:0]	R/W		0x01
0x75	bit[2:0]	To_Low_Limit[10:8]	R/W	4	0x04
0x76		Ta_Hysteresis[7:0]	R/W		0x00
0x77		To_Hysteresis[7:0]	R/W		0x00
0x7D		Host_SW_Reset[7:0]	W		0x00

Table 30. Register Bank 1

Address	Bit Field	Name	Access	Bit Field Default Value	Register Default Value
0x50	Bit[6:0]	Emissivity[6:0]	R/W	96	0x60
0x51		Decay[7:0]	R/W		0x00
0x5B	Bit[6:5]	FIR_Stage[2:0]	R/W	0	0x10
	Bit[4:3]	Filter_Mode[1:0]	R/W	2	
	Bit[2:0]	IIR_level[2:0]	R/W	0	

8.2 Register Description

8.2.1 Software Reset Register

Register Name	Host_SW_Reset[7:0]		
Bank	0x00	Address	0x7D
Access	W	Default Value	0x00
Description	Set value to 0x5A for cold reset (all digital functions will be reset). Set value to 0x9A for warm reset (all digital functions will be reset except the registers).		

8.2.2 Product ID Register

Register Name	PartID[7:0]		
Bank	0x00	Address	0x00
Access	R	Default Value	0x71
Register Name	PartID[15:8]		
Bank	0x00	Address	0x01
Access	R	Default Value	0x02
Description	Part ID = 0x0271		

8.2.3 Bank Switch Register

Register Name	Cmd_Bank_Sel[7:0]		
Bank	-	Address	0x7F
Access	R/W	Default Value	0x00
Bit Field	Name	Default Value	Description
Description	Bank0: Write value 0x00 to switching address mapping register to bank0. Bank1: Write value 0x01 to switching address mapping register to bank1.		

8.2.4 System Mode Switching Registers

Register Name	Alert_Mode[1:0]		
Bank	0x00	Address	0x03
Access	R/W	Default Value	0x00
Bit Field	Name	Default Value	Description
1:0	Alert_Mode[1:0]	0	Alert mode switching to change alert signal behavior. 0: Normal Mode 1: Interrupt Mode 2: Compare Mode

Register Name	One-Shot		
Bank	0x00	Address	0x26
Access	R/W	Default Value	0x00
Bit Field	Name	Default Value	Description
1	One-Shot	0	The difference possible operation modes are: 0: Burst Mode (switching to <i>Alert_Mode</i> option) 1: One-Shot Mode

8.2.5 Limit Window and Hysteresis Registers

Register Name	Ta_High_Limit[7:0]		
Bank	0x00	Address	0x6E
Access	R/W	Default Value	0xFF
Register Name	Ta_High_Limit[10:8]		
Bank	0x00	Address	0x6F
Access	R/W	Default Value	0x03
Bit Field	Name	Default Value	Description
2:0	Ta_High_Limit[10:8]	3	Bit[10:8] of the Ta_High_Limit[10:0]
Description	Ta High Limit value setting. Upper threshold = High Limit. The data format is using 11 bits 2's complement. The setting scale is 0.5°C/LSB.		

Register Name	Ta_Low_Limit[7:0]		
Bank	0x00	Address	0x70
Access	R/W	Default Value	0x01
Register Name	Ta_Low_Limit[10:8]		
Bank	0x00	Address	0x71
Access	R/W	Default Value	0x04
Bit Field	Name	Default Value	Description
2:0	Ta_Low_Limit[10:8]	4	Bit[10:8] of the Ta_Low_Limit[10:0]
Description	Ta Low Limit value setting. Lower threshold = Lower Limit. The data format is using 11 bits 2's complement. The setting scale is 0.5°C/LSB.		

Register Name	To_High_Limit[7:0]		
Bank	0x00	Address	0x72
Access	R/W	Default Value	0xFF
Register Name	To_High_Limit[10:8]		
Bank	0x00	Address	0x73
Access	R/W	Default Value	0x03
Bit Field	Name	Default Value	Description
2:0	To_High_Limit[10:8]	3	Bit[10:8] of the To_High_Limit[10:0]
Description	To High Limit value setting. Upper threshold = High Limit. The data format is using 11 bits 2's complement. The setting scale is 0.5°C/LSB.		

Register Name	To_Low_Limit[7:0]		
Bank	0x00	Address	0x74
Access	R/W	Default Value	0x01
Register Name	To_Low_Limit[10:8]		
Bank	0x00	Address	0x75
Access	R/W	Default Value	0x04
Bit Field	Name	Default Value	Description
2:0	To_Low_Limit[10:8]	3	Bit[10:8] of the To_Low_Limit[10:0]
Description	To Low Limit value setting. Lower threshold = Lower Limit. The data format is using 11 bits 2's complement. The setting scale is 0.5°C/LSB.		

Register Name	Ta_Hysteresis[7:0]		
Bank	0x00	Address	0x76
Access	R/W	Default Value	0x00
Description	Ta Hysteresis setting of ambient temperature. The data format is using 8 bits. The setting scale is 0.5°C/LSB.		

Register Name	To_Hysteresis[7:0]		
Bank	0x00	Address	0x77
Access	R/W	Default Value	0x00
Description	To Hysteresis setting of object temperature. The data format is using 8 bits. The setting scale is 0.5°C/LSB.		

8.2.6 ADC Raw Data Registers

Register Name	DSP_Ta_Data[7:0]		
Bank	0x00	Address	0x06
Access	R	Default Value	0x00
Register Name	DSP_Ta_Data[15:8]		
Bank	0x00	Address	0x07
Access	R	Default Value	0x00
Description	Ta ADC code. The data format is using 16 bits 2's complement.		

Register Name	DSP_To_Data[7:0]		
Bank	0x00	Address	0x08
Access	R	Default Value	0x00
Register Name	DSP_To_Data[15:8]		
Bank	0x00	Address	0x09
Access	R	Default Value	0x00
Description	To ADC code. The data format is using 16 bits 2's complement.		

8.2.7 Temperature Data Registers

Register Name	CAL_Ta_Data[7:0]		
Bank	0x00	Address	0x0A
Access	R	Default Value	0x00
Register Name	CAL_Ta_Data[15:8]		
Bank	0x00	Address	0x0B
Access	R	Default Value	0x00
Description	Ta ambient temperature. The data format is using 16 bits 2's complement. The output scale is 0.03125°C/LSB.		

Register Name	CAL_To_Data[7:0]		
Bank	0x00	Address	0x0C
Access	R	Default Value	0x00
Register Name	CAL_To_Data[15:8]		
Bank	0x00	Address	0x0D
Access	R	Default Value	0x00
Description	To object temperature. The data format is using 16 bits 2's complement. The output scale is 0.03125°C/LSB.		

8.2.8 Status Registers

The status flag is updated for each conversion to indicate sensor status and temperature data event.

Register Name	Status Register		
Bank	0x00	Address	0x05
Access	R	Default Value	0x00
Description	Indicate sensor status and temperature data event.		
Bit Field	Name	Default Value	Description
7	Data_Overflow_Flag	0	This bit is set to 1 if the result in conversion data is over flow and the conversion data is invalid.
6	OTP_Load_Done	0	OTP Load Done flag. This bit is set to 1 if the calibration coefficients load from internal memory to system is completed.
5	Host_Clear_Interrupt	0	This bit is set to 1 if the host needs to clear all alert signals. Refer to Section 7.4.1 .
4	To_Low_Limit_Flag	0	This bit is set to 1 if the result in the object temperature register is less than the value in the object temperature Low-Limit register.
3	To_High_Limit_Flag	0	This bit is set to 1 if the result in the object temperature register exceeds the value in the object temperature High-Limit register.
2	Ta_Low_Limit_Flag	0	This bit is set to 1 if the result in the ambient temperature register is less than the value in the object temperature Low-Limit register.
1	Ta_High_Limit_Flag	0	This bit is set to 1 if the result in the ambient temperature register exceeds the value in the ambient temperature High-Limit register.
0	Alert_Flag	0	Set to 1 if the conversion is done.

8.2.9 Sampling Number Registers

Register Name	Burst_Num_Sel[7:0]		
Bank	0x00	Address	0x20
Access	R/W	Default Value	0x9C
Bit Field	Name	Default Value	Description
7:4	Ta_Burst_Num_Sel[3:0]	9	Local Ambient sensor's ADC samples (M) and sampling number is 2^M .
3:0	To_Burst_Num_Sel[3:0]	12	Object sensor's ADC samples (N) and sampling number is 2^N .

8.2.10 Report Rate Registers

Register Name	Burst_Freq_Sel[7:0]		
Bank	0x00	Address	0x21
Access	R/W	Default Value	0x38
Register Name	Burst_Freq_Sel[15:8]		
Bank	0x00	Address	0x22
Access	R/W	Default Value	0x01
Register Name	Burst_Freq_Sel[19:16]		
Bank	0x00	Address	0x23
Access	R/W	Default Value	0x00
Bit Field	Name	Default Value	Description
3:0	Burst_Freq_Sel[19:16]	0	Bit[19:16] of the Burst_Freq_Sel[19:0]
Description	Frame time (sec) = $(1 \div 20k) \times \text{Burst_Freq_Sel}[19:0] \times 16$ = 0.2496 (sec)		
	Report rate $\approx 1 \div \text{Frame time}$ = 4Hz (default)		

8.2.11 Filter Registers

Register Name	Filter Register[7:0]		
Bank	0x01	Address	0x5B
Access	R/W	Default Value	0x10
Bit Field	Name	Default Value	Description
6:5	FIR_Stage[1:0]	0	0: 1 frame average (default setting) 1: 2 frame average 2: 4 frame average 3: 8 frame average
4:3	Filter_Mode[1:0]	2	0: IIR Mode 1: Moving Average Mode 2: Normal Average Mode (default setting)
2:0	IIR level[2:0]	0	0: $\text{Frame}(n-1) \times 0 + \text{Frame}(n) \times 1$ (default setting) 1: $\text{Frame}(n-1) \times 0.125 + \text{Frame}(n) \times 0.875$ 2: $\text{Frame}(n-1) \times 0.25 + \text{Frame}(n) \times 0.75$ 3: $\text{Frame}(n-1) \times 0.375 + \text{Frame}(n) \times 0.625$ 4: $\text{Frame}(n-1) \times 0.5 + \text{Frame}(n) \times 0.5$ 5: $\text{Frame}(n-1) \times 0.625 + \text{Frame}(n) \times 0.375$ 6: $\text{Frame}(n-1) \times 0.75 + \text{Frame}(n) \times 0.25$ 7: $\text{Frame}(n-1) \times 0.875 + \text{Frame}(n) \times 0.125$

8.2.12 Emissivity Register

Register Name	Emissivity[6:0]		
Bank	0x01	Address	0x50
Access	R/W	Default Value	0x60
Bit Field	Name	Default Value	Description
6:0	Emissivity[6:0]	96	The default setting is 0.96 (96%) of sensor emissivity coefficient. Emissivity % = Emissivity[6:0]

8.2.13 Decay Register

Register Name	Decay[7:0]		
Bank	0x01	Address	0x51
Access	R/W	Default Value	0x00
Description	Cover decay compensation. Cover decay % = Decay[7:0]. The setting scale is 0.5°C/LSB.		

8.2.14 ALERT Open Drain Register

Register Name	ALERT_Open_Drain		
Bank	0x00	Address	0x2F
Access	R/W	Default Value	0x00
Bit Field	Name	Default Value	Description
0	ALERT_Open_Drain	0	0: Disable alert pin open drain behavior 1: Enable alert pin open drain behavior

Revision History

Revision Number	Date	Description
0.5	20 Nov 2024	Initial release
0.51	16 May 2025	Correct signal pins description table Modify operation temperature

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